



LPDDR5X SDRAM

**MT62F3G32D4, MT62F6G32D8, MT62F1536M32D2,
MT62F1536M64D4, MT62F3G64D8**

Features

- **Architecture**
 - 21.3 GB/s maximum bandwidth per channel
 - Frequency range: 1333–5 MHz (data rate range per pin: 10,667–40 Mb/s with WCK:CK = 4:1)
 - Selectable CKR (WCK:CK = 2:1 or 4:1)
- **LPDDR5X/LPDDR5 data interface**
 - Single x16 channel/die
 - Double-data-rate command/address entry
 - Differential command clocks (CK_t/CK_c) for high-speed operation
 - Differential data clocks (WCK_t/WCK_c)
 - Optional differential read strobe (RDQS_t/RDQS_c)
 - 16n-bit or 32n-bit prefetch architecture
 - Command-selectable burst lengths (BL = 16 or 32) in bank group or 16-bank modes
 - Background ZQ calibration/command-based ZQ calibration
 - Optional link protection (link ECC)
 - Partial-array self refresh (PASR) and partial-array auto refresh (PARC) with segment mask
- **Ultra-low-voltage core and I/O power supplies**
 - V_{DD1} = 1.70–1.95V; 1.80V TYP
 - V_{DD2H} = 1.01–1.12V; 1.05V TYP
 - V_{DD2L} = V_{DD2H} or 0.87–0.97V; 0.90V TYP
 - V_{DDQ} = 0.50V or 0.45V¹ TYP; 0.30V TYP (ODT off only)
- **I/O characteristics**
 - Interface-LVSTL 0.5/0.3
 - I/O type: Low-swing single-ended, V_{SS} terminated
 - V_{OH}-compensated output drive
 - Programmable V_{SS} on-die termination (ODT)
 - Non target ODT support
 - DVFSQ support
- **Low power features**
 - Enhanced DVFSQ: Enhanced Dynamic voltage frequency scaling core
 - Single-ended CK, single-ended WCK and single-ended RDQS
 - Data copy
 - Write X

Options

- **Operating voltage**
 - V_{DD1}/V_{DD2H}/V_{DD2L}/V_{DDQ}/V_{DDQ} (ODT off only): 1.80V/1.05V/V_{DD2H} or 0.90V/0.50V or 0.45V¹/0.30V
- **Array configuration**
 - 1536Mb x 32 (1536Mb16 x 1 die x 2Ch x 1R)
 - 3Gb x 32 (1536Mb16 x 1 die x 2Ch x 2R)
 - 6Gb x 32 (3Gb8 x 2 die x 2Ch x 2R)
 - 1536Mb x 64 (1536Mb16 x 1 die x 4Ch x 1R)
 - 3Gb x 64 (1536Mb16 x 1 die x 4Ch x 2R)
- **Device configuration**
 - 2 die in package (1536Mb16 x 2 die)
 - 4 die in package (1536Mb16 x 4 die)
 - 8 die in package (1536Mb16 x 8 die)
 - 8 die in package (3Gb8 x 8 die)
- **FBGA RoHS-compliant, "green" package**
 - 315-ball TFBGA
 - 315-ball LFBGA
 - 441-ball TFBGA
 - 561-ball TFBGA
- **Speed grade, cycle time (^tWCK)**
 - 10,667 Mb/s
 - 9600 Mb/s
- **Operating temperature**
 - -25°C ≤ T_C ≤ +85°C
 - -40°C ≤ T_C ≤ +95°C
- **Revision**

Marking

F

1536M32
3G32
6G32
1536M64
3G64

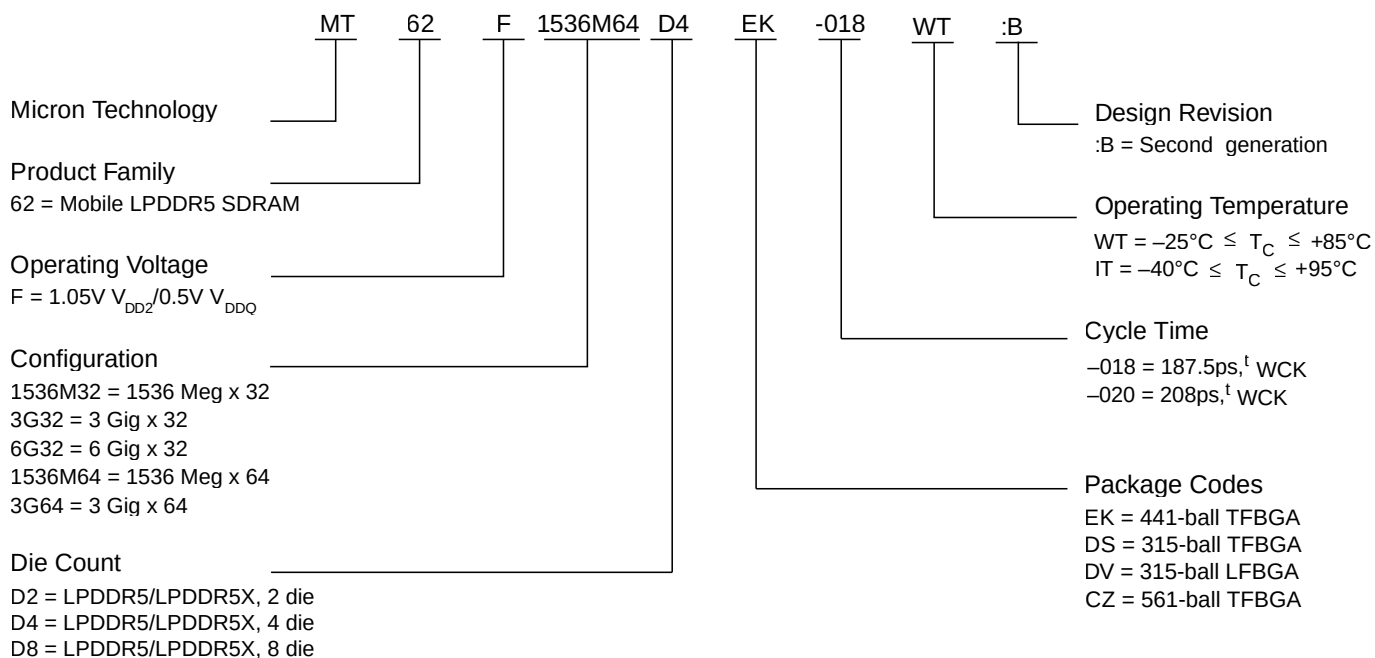
D2
D4
D8
D8

DS
DV
EK
CZ

-018
-020

WT
IT
:B

Note: 1. V_{DDQ} = 0.45V (TYP) support on all packages up to 7500 Mb/s per-pin.


Y6CP LPDDR5X SDRAM
Features
Part Number Ordering Information
Figure 1: Part Number Chart

Table 1: Part Number List

Part Number	Total Density	Data Rate per Pin
MT62F1536M32D2DS-018 WT:B	6GB (48Gb)	10,667 Mb/s
MT62F3G32D4DS-018 WT:B	12GB (96Gb)	
MT62F6G32D8DV-018 WT:B	24GB (192Gb)	
MT62F1536M64D4EK-018 WT:B	12GB (96Gb)	
MT62F3G64D8EK-018 WT:B	24GB (192Gb)	
MT62F1536M64D4CZ-018 WT:B	12GB (96Gb)	
MT62F3G64D8CZ-018 WT:B	24GB (192Gb)	
MT62F1536M32D2DS-020 WT:B	6GB (48Gb)	9600 Mb/s
MT62F1536M32D2DS-020 IT:B		
MT62F3G32D4DS-020 WT:B	12GB (96Gb)	
MT62F3G32D4DS-020 IT:B		
MT62F6G32D8DV-020 WT:B	24GB (192Gb)	
MT62F6G32D8DV-020 IT:B		
MT62F1536M64D4EK-020 WT:B	12GB (96Gb)	
MT62F3G64D8EK-020 WT:B	24GB (192Gb)	
MT62F1536M64D4CZ-020 WT:B	12GB (96Gb)	
MT62F3G64D8CZ-020 WT:B	24GB (192Gb)	



FBGA Part Marking Decoder

Due to space limitations, FBGA-packaged components have an abbreviated part marking that is different from the part number. Micron's FBGA part marking decoder is available at www.micron.com/decoder.

LPDDR5/LPDDR5X Data Sheet List

For general LPDDR5/LPDDR5X specifications, please refer to the data sheets below.

- General LPDDR5/LPDDR5X Specifications 1: Mode Registers
- General LPDDR5/LPDDR5X Specifications 2: AC/DC and Interface Specifications
- General LPDDR5/LPDDR5X Specifications 3: Features and Functionalities



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Important Notes and Warnings

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General Notes

Throughout the data sheet, figures and text refer to DQs as DQ. DQ should be interpreted as any or all DQs collectively, unless specifically stated otherwise.

RDQS, CK, and WCK should be interpreted as RDQS_t, RDQS_c, CK_t, CK_c, and WCK_t, WCK_c respectively unless specifically stated otherwise. CA includes all CA pins used for a given density.

In timing diagrams, CMD is used as an indicator only. Actual signals occur on CA[6:0].

V_{REF} indicates $V_{REF(CA)}$, $V_{REF(CS)}$, and $V_{REF(DQ)}$.

Complete functionality is described throughout the entire document. Any page or diagram may have been simplified to convey a topic and may not be inclusive of all requirements.

Any content defined in this device ID specific data sheet document takes precedence over similar content defined in the general core LPDDR5/LPDDR5X SDRAM data sheet document(s).

Any functionality not specifically stated herein is considered undefined, illegal, is not supported, and will result in unknown operation.



Device Configuration

Table 2: Die Organization in the Package (x32)

Die Organization	1536M32 (48 Gb/package)	3G32 (96 Gb/package)	6G32 (192 Gb/package)
Channel A, rank0	x16 mode x 1 die	–	–
Channel B, rank0	x16 mode x 1 die	–	–
Channel A, rank 0	–	x16 mode x 1 die	–
Channel B, rank 0	–	x16 mode x 1 die	–
Channel A, rank 1	–	x16 mode x 1 die	–
Channel B, rank 1	–	x16 mode x 1 die	–
Channel A, rank 0 DQ[7:0]	–	–	x8 mode x 1 die
Channel A, rank 1 DQ[7:0]	–	–	x8 mode x 1 die
Channel B, rank 0 DQ[7:0]	–	–	x8 mode x 1 die
Channel B, rank 1 DQ[7:0]	–	–	x8 mode x 1 die
Channel A, rank 0 DQ[15:8]	–	–	x8 mode x 1 die
Channel A, rank 1 DQ[15:8]	–	–	x8 mode x 1 die
Channel B, rank 0 DQ[15:8]	–	–	x8 mode x 1 die
Channel B, rank 1 DQ[15:8]	–	–	x8 mode x 1 die

Note: 1. Refer to the Package Block Diagram section in this data sheet.

Table 3: Die Organization in the Package (x64)

Die Organization	1536M64 (96 Gb/package)	3G64 (192 Gb/package)
Channel A, rank 0	x16 mode x 1 die	x16 mode x 1 die
Channel B, rank 0	x16 mode x 1 die	x16 mode x 1 die
Channel C, rank 0	x16 mode x 1 die	x16 mode x 1 die
Channel D, rank 0	x16 mode x 1 die	x16 mode x 1 die
Channel A, rank 1	–	x16 mode x 1 die
Channel B, rank 1	–	x16 mode x 1 die
Channel C, rank 1	–	x16 mode x 1 die
Channel D, rank 1	–	x16 mode x 1 die

Note: 1. Refer to the Package Block Diagram section in this data sheet.


Y6CP LPDDR5X SDRAM
Device Configuration
Table 4: Die Addressing

Description	1536M32 (48 Gb/pkg), 3G32 (96 Gb/pkg), 1536M64 (96 Gb/pkg), 3G64 (192 Gb/pkg)		6G32 (192 Gb/package)	
	BG mode	16B mode	BG mode	16B mode
Density per die	24Gb		24Gb	
Bits	25,769,803,776		25,769,803,776	
Configuration	96Mb × 16 DQ × 4 banks × 4BG	96Mb × 16 DQ × 16 banks	192Mb × 8 DQ × 4 banks × 4BG	192Mb × 8 DQ × 16 banks
Number of banks	4	16	4	16
Number of bank groups	4	1	4	1
Array prefetch bits	256	256	128	128
Rows per bank	98,304		196,608	
Columns	64		64	
Page size (bytes)	2048	2048	1024	1024
Native burst length	16	16	16	16
Number of I/Os	16		8	
Bank address	BA[1:0]	BA[3:0]	BA[1:0]	BA[3:0]
Bank group address	BG[1:0]	–	BG[1:0]	–
Row address	R[16:0] (R15 = 0 when R16 = 1)		R[17:0] (R16 = 0 when R17 = 1)	
Column address	C[5:0]		C[5:0]	
Burst address	B[3:0]	B[3:0]	B[3:0]	B[3:0]
Burst starting address boundary	128-bit		128-bit	

Notes: 1. Refer to the SDRAM Addressing section in General LPDDR5/LPDDR5X Specifications 3.
 2. Refer to the Speed Grades and Effective Burst Length in General LPDDR5/LPDDR5X Specifications 3.



Y6CP LPDDR5X SDRAM Refresh Requirement Parameters

Refresh Requirement Parameters

Table 5: Refresh Requirement Parameters

Parameter	Symbol	24Gb Die	Unit
		BG and 16B Mode	
REFRESH cycle time (all banks)	t_{RFCab}	380	ns
REFRESH cycle time (per bank)	t_{RFCpb}	190	ns
Per bank refresh to per bank refresh time (different bank)	$t_{PBR2PBR}$	90	ns
Per bank refresh to ACTIVATE command time (different bank)	$t_{PBR2ACT}$	7.5	ns

Note: 1. This table only describes refresh parameters that are density dependent. Refer to Refresh Requirement section in General LPDDR5/LPDDR5X Specifications 3 for all refresh parameters.


Y6CP LPDDR5X SDRAM
CS Rx Relaxed Specification

CS Rx Relaxed Specification

Table 6: CS Rx Specifications

Item	Symbol	Min/ Max	CK Frequency (MHz)													Unit	Note
			67 ⁶	133	200	266	344	400	467	533	600	688	750	800	937.5		
Rx mask																	
CS Rx mask height	vCSIVW	Max	155													mV	2, 7
CS Rx mask width at V _{REF} CS	^t CSIVW1	Max	0.3													UI	1
CS Rx mask width at vCSIVW	^t CSIVW2	Max	0.22													UI	1
Rx single pulse																	
CS Rx pulse amplitude	vCSIHL_AC	Min	240													mV	3
CS reference voltage	vREFCS		V _{DD2H} /3													mV	
CS Rx pulse width	^t CSIPW	Min	0.6													UI	
Power down																	
CS V _{IL} during power down/ deep sleep	V _{ILPD}	Max	130													mV	4
CS V _{IH} during power down/ deep sleep	V _{IHPD}	Min	550													mV	5
		Max	V _{DD2H} + 200														

- Notes: 1. CS Rx mask voltage and timing parameters at the pin include temperature drift and voltage AC noise impact based on Z(f) specification at a fixed temperature on the package. The voltage supply noise must comply to the component min/max DC operating conditions.
2. CS single-pulse signal amplitude into the receiver must meet or exceed vCSIHL_AC at any point over the total UI; No timing requirement above a certain level. vCSIHL_AC is the peak-to-peak voltage centered around V_{REF} CS such that vCSIHL_AC/2 min has to be met both above and below V_{REF} CS.
3. vCSIHL_AC does not have to be met when no transitions are occurring.
4. The input voltage presented to the CS Rx pin during power down should be 0V nominally to minimize leakage current.
5. V_{IHPD} is only applied for POWER DOWN and DEEP SLEEP EXIT operations.
6. The Rx voltage and absolute timing requirements apply for all CS operating frequencies at or below 67 for all speed bins. For example, ^tCSIVW1 (ns) = 4.477ns at or below 67 MHz CK frequency.
7. Measured at die pad.



Y6CP LPDDR5X SDRAM

DQ Rx Mask and Single Input Pulse Relaxed Specifications

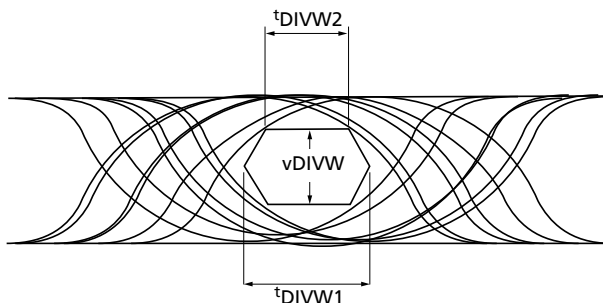
DQ Rx Mask and Single Input Pulse Relaxed Specifications

Table 7: DQ Rx Mask and Single Input Pulse Specifications

Item	Symbol	Min/ Max	WCK Frequency (MHz)			Unit	Notes
			3750	4266	4800		
DQ Rx mask height	vDIVW	Max	70			mV	1, 2, 3
DQ Rx mask width at $V_{REF(DQ)}$	tDIVW1	Max	0.35			UI	1, 3
DQ Rx mask width at vDIVW	tDIVW2	Max	0.18			UI	1, 3
DQ Rx pulse width at $V_{REF(DQ)}$	tDIPW1	Min	0.51			UI	4
DQ Rx pulse reference	tDIPW2	Min	0.26			UI	4
DQ Rx pulse width at $V_{REF(DQ)} \pm vDIVW/2$	tDIHL	Min	0.24			UI	4
DQ Rx pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP1	Min	70			mV	
	vDILP1	Max	-70			mV	
DQ Rx early pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP2	Min	55			mV	
	vDILP2	Max	-55			mV	
DQ Vref	VrefDQ	Max	180			mV	
		Min	75			mV	
DQ to DQ Offset	tDQ2DQ	Max	30			ps	5

- Notes: 1. Data Rx mask voltage and timing parameters are applied per pin and include the DRAM DQ-to-DQS voltage AC noise impact for frequencies greater than TBD MHz and maximum voltage of TBD mV peak-to-peak from DC-TBD MHz at a fixed temperature on the package. The voltage supply noise must comply with the component min/max DC operating conditions.
2. Rx mask voltage vDIVW must be centered around $V_{REF(DQ)}$.
3. Mask specifications (tDIVW1 , tDIVW2 , and vDIVW) are the same regardless DFE enabled or disabled. DFE co-efficient can be included as part of input waveform amplitude when DFE is enabled.
4. UI = ${}^tWCK/2$, programed V_{REF} , is defined as a percentage of MR14/15 code x V_{DDQ} .
5. DQ to DQ offset defined within byte from DRAM pin to internal latch for a given component.

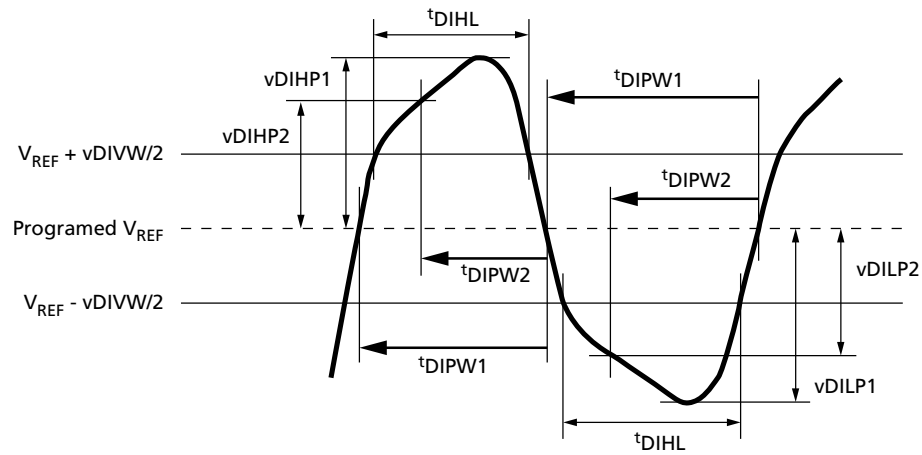
Figure 2: DQ Rx Mask Definition





Y6CP LPDDR5X SDRAM DQ Rx Mask and Single Input Pulse Relaxed Specifications

Figure 3: LPDDR5X DQ Single Pulse Definition



Note: 1. $v_{DIHL_AC/2} = v_{DIHP1} = v_{DILP1}$. Programed $V_{REF(DQ)}$ is defined as percentage of MR14/15 code * V_{DDQ} .



Y6CP LPDDR5X SDRAM DQ Rx Mask and Single Input Pulse Relaxed Specifications 3200 MHz

DQ Rx Mask and Single Input Pulse Relaxed Specifications 3200 MHz

Table 8: DQ, DMI, Parity and DBI Rx Mask and Single Input Pulse Specifications

Item	Symbol	Min/Max	WCK Frequency (MHz)	Unit	Notes
			3200		
DQ Rx mask height	vDIVW	Max	70	mV	1, 2, 3, 6
DQ Rx mask width at $V_{REF(DQ)}$	t^{DIVW1}	Max	0.35	UI	1, 3
DQ Rx mask width at vDIVW	t^{DIVW2}	Max	0.18	UI	1, 3
DQ Rx pulse width at $V_{REF(DQ)}$	t^{DIPW1}	Min	0.51	UI	4, 6
DQ Rx pulse reference	t^{DIPW2}	Min	0.26	UI	4
DQ Rx pulse width at $V_{REF(DQ)} \pm vDIVW/2$	t^{DIHL}	Min	0.24	UI	4, 6
DQ Rx pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP1	Min	70	mV	
	vDILP1	Max	-70	mV	
DQ Rx early pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP2	Min	55	mV	
	vDILP2	Max	-55	mV	
DQ V_{REF}	$V_{REF(DQ)}$	Min	75	mV	6
		Max	180	mV	6
DQ to DQ offset	t^{DQ2DQ}	Max	30	ps	5

- Notes: 1. Data Rx mask voltage and timing parameters are applied per pin and include the DRAM DQ-to-DQS voltage AC noise impact for frequencies greater than TBD MHz and maximum voltage of TBD mV peak-to-peak from DC-TBD MHz at a fixed temperature on the package. The voltage supply noise must comply with the component min/max DC operating conditions.
2. Rx mask voltage vDIVW must be centered around $V_{REF(DQ)}$.
3. Mask specifications (t^{DIVW1} , t^{DIVW2} , and vDIVW) are the same regardless DFE enabled or disabled. DFE co-efficient can be included as part of input waveform amplitude when DFE is enabled.
4. UI = $t^{WCK}/2$, programed V_{REF} , is defined as a percentage of MR14/15 code x V_{DDQ} .
5. DQ to DQ offset defined within byte from DRAM pin to DRAM internal latch for a given component.
6. When vDIVW is 70mV or more and less than 80mV.



16-Bank Extended Frequency

16-Bank Extended Frequency Description

Micron's LPDDR5 SDRAM device which are marked with 8533 Mb/s per-pin (-023) capability can support frequency set points using 2:1 WCK to CK ratio and above the 1600 MHz limit. Shown in the table below is specification requirements when considering 1867 MHz and 2133 MHz frequencies. This support is not available on 1-alpha process node and earlier Device ID's, this feature can expand capabilities of your system.

Table 9: 16-Bank Operation¹ @ 4.3 Gb/s

Parameter	3733 Mb/s per-pin	Comments	4267 Mb/s per-pin	Comments
RL Set 1/2/3	20/22/24	MR2[3:0] = 1100	23/25/26	MR2[3:0] = 1101
nRBTP	6		6	
WL Set A/B	11/19	MR1[7:4] = 1100	12/22	MR1[7:4] = 1101
nWR x16/x8	32/34	MR2[7:4] = 1100	37/39	MR2[7:4] = 1101
^t WCKPRE_static ²	5	Follows 7500 data rate	6	Follows 8533 data rate
^t WCKPRE_WR_toggle	4	Follows 3200 data rate	4	Follows 3200 data rate
^t WCKPRE_RD_toggle	10	Follows 3200 data rate	10	Follows 3200 data rate
^t WCKENL_FS	2	Follows 3200 data rate	2	Follows 3200 data rate
^t WCKPRE_toggle_FS ³	14	Minimum	16	Minimum
ODTLon	WL - 4	Follows 3200 data rate	WL - 4	Follows 3200 data rate
ODTLoff ⁴	WL + 5 (BL16), WL + 9 (BL32)	Follows 3200	WL + 5 (BL16), WL + 9 (BL32)	Follows 3200 data rate
ODTLoff_RD_DQ	RL - 4	Follows 3200 data rate	RL - 4	Follows 3200 data rate
ODTLoff_RD_RDQS	RL - 6	MR10 OP[1, 5:4] = 000, 001, 010	RL - 6	MR10 OP[1, 5:4] = 000, 001, 010
	RL - 8	MR10 OP[1, 5:4] = 011, 1XX	RL - 8	MR10 OP[1, 5:4] = 011, 1XX
ODTLon_RD_DQ ⁵	RL + 6 (BL16), RL + 10 (BL32)		RL + 6 (BL16), RL + 10 (BL32)	
ODTLon_RD_RDQS ⁶	RL + 8 (BL16), RL + 12 (BL32)		RL + 8 (BL16), RL + 12 (BL32)	

Notes: 1. Restricted to WCK:CK 2:1 ratio, WCK:CK 4:1 ratio not supported; Link ECC is disabled, and DVFS/E-DVFS is disabled.

2. Common for WR/RD/FS.

3. Minimum 1 + RL - (^tWCKENL_FS + ^tWCKPRE_Static) when CAS to RD is 1CK.

4. (WL + BL/n_min + RU(^tWCK2DQI(max)/^tCK)).

5. (RL + BL/n_min + RU(^tWCK2DQO(max)/^tCK)).

6. (RL + BL/n_min + RU(^tWCK2DQO(max)/^tCK)) + 2.

7. 8533 MT/s or higher data rated parts support 16-bank mode up to 4267 MT/s with restrictions, all other speed rated parts greater than 3200 MT/s per-pin and less than 8533 MT/s per-pin are restricted from the extended 16B mode capabilities. Refer to the above table for constraints for operating in 16-bank mode when data rate is greater than 3200 MT/s per-pin and equal to or less than 4267 MT/s per-pin.

8. When operating in the 16-bank mode, the FSP procedure must be used when setting the data rate above 3.2 Gb/s per-pin even when setting for the first time. For example, if the device is powered-up and initialized for 4266 Gb/s per-pin timing at the initial operation start, the FSP procedure must still be used. See [Figure 4: 16B and BG Data Rate Change Flow Chart Showing Extended Capabilities](#) on page 17 for details of FSP usage.

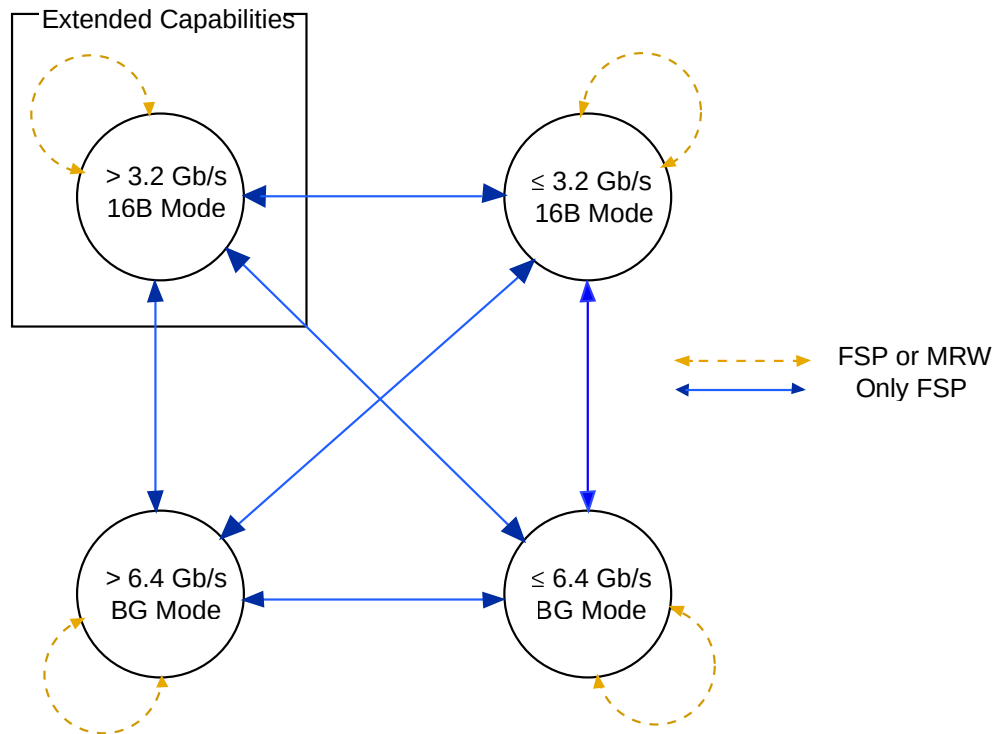


16B/BG Data Rate Change Diagram With FSP Requirements

The following flow chart diagram shows the needed 16-bank (16B) and bank group (BG) data rate change details and how the frequency set protocol (FSP) supports the extended frequency range.

The diagram below represents the device extended capabilities which are extended to 4267 Mb/s per-pin data rate in 16B mode.

Figure 4: 16B and BG Data Rate Change Flow Chart Showing Extended Capabilities

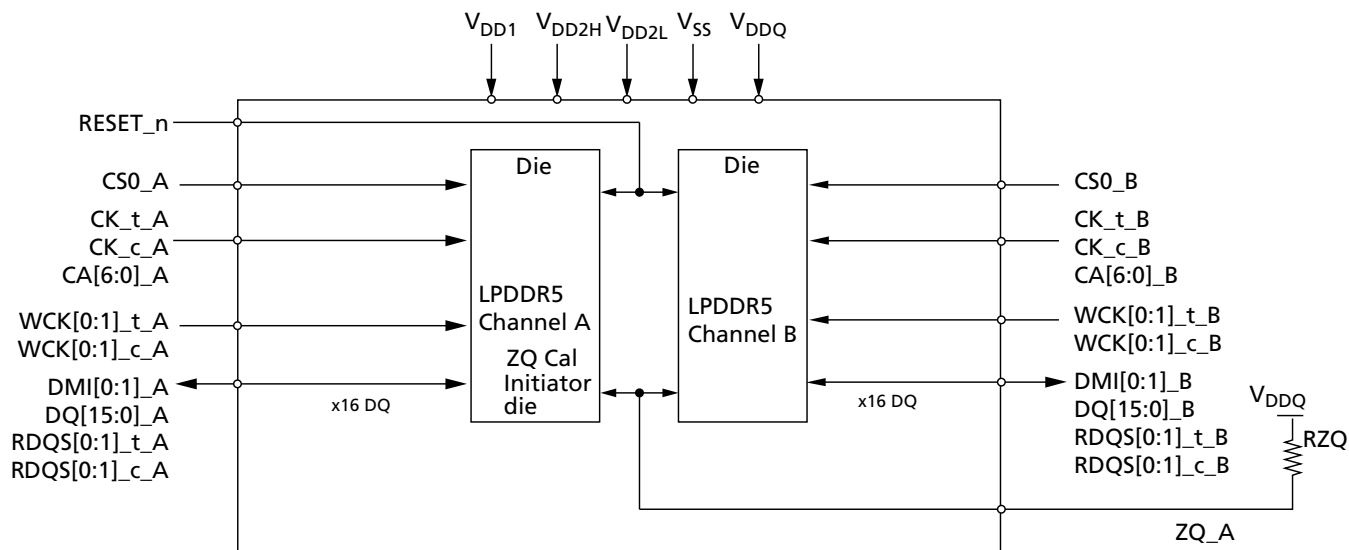




Package Block Diagrams

Dual Die, Dual Channel, Single Rank

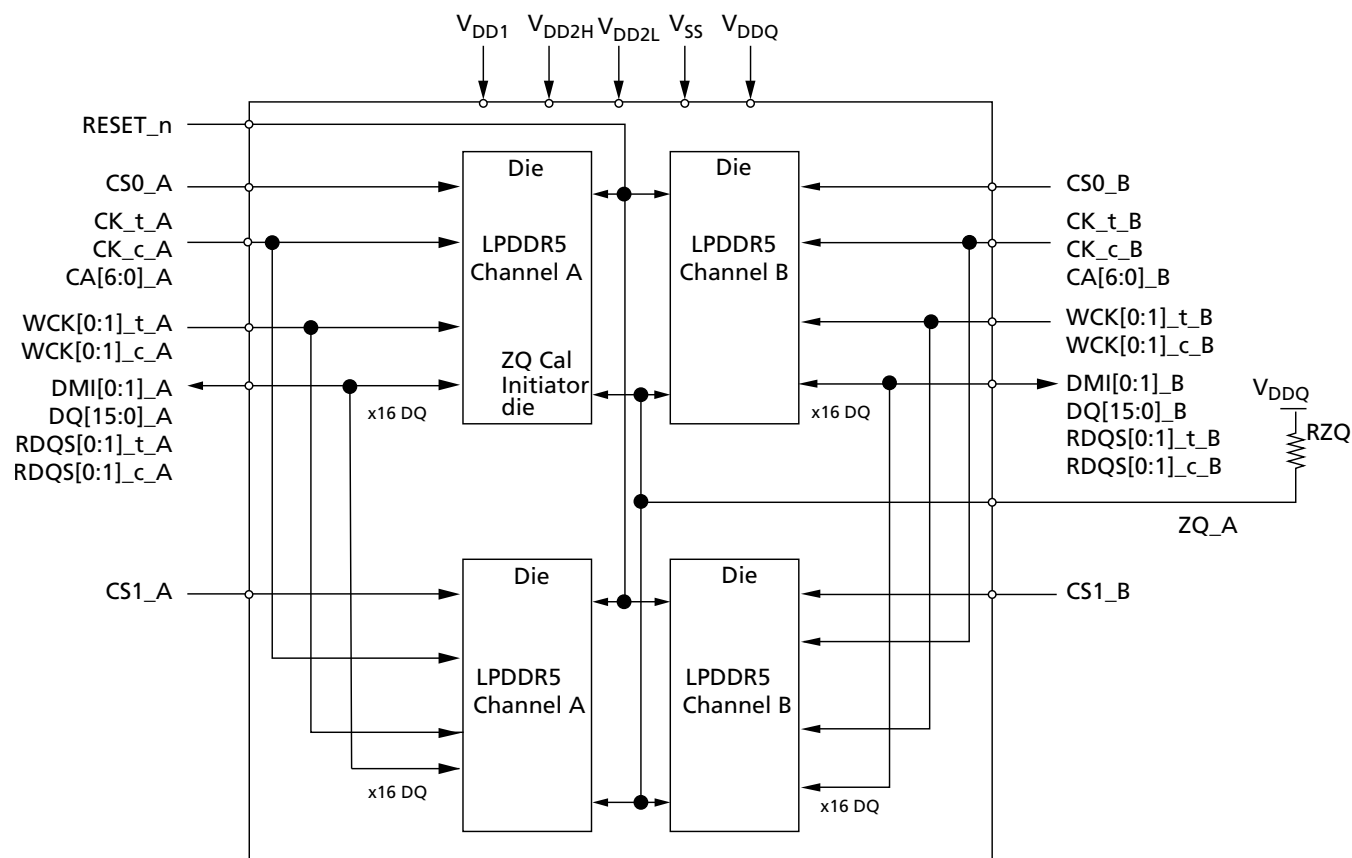
Figure 5: Dual-Die, Dual-Channel, Single-Rank Package Block Diagram





Quad Die, Dual Channel, Dual Rank

Figure 6: Quad-Die, Dual-Channel, Dual-Rank Package Block Diagram

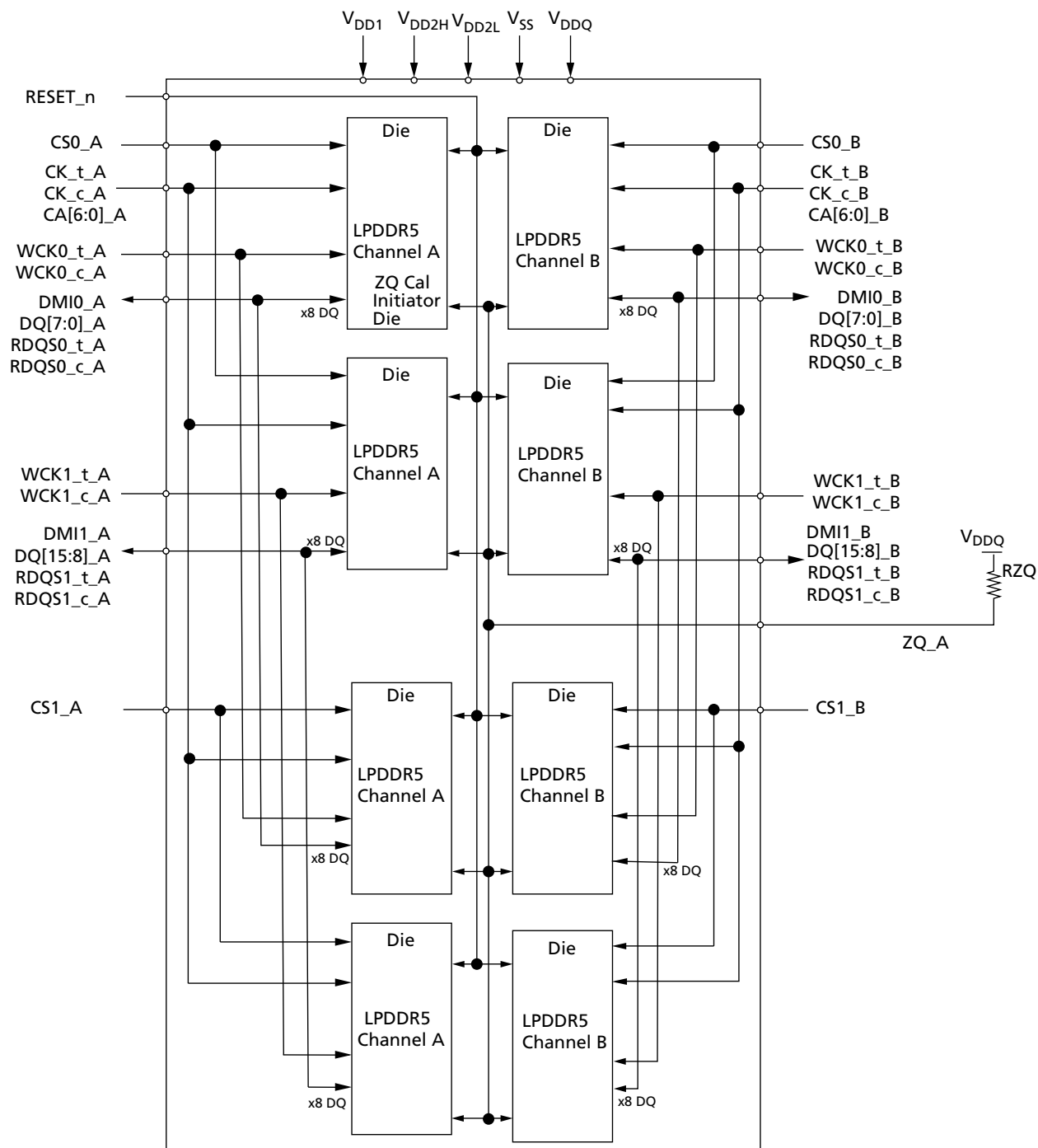




Y6CP LPDDR5X SDRAM Package Block Diagrams

Eight Die, Dual Channel, Dual Rank

Figure 7: Eight-Die, Dual-Channel, Dual-Rank Package Block Diagram

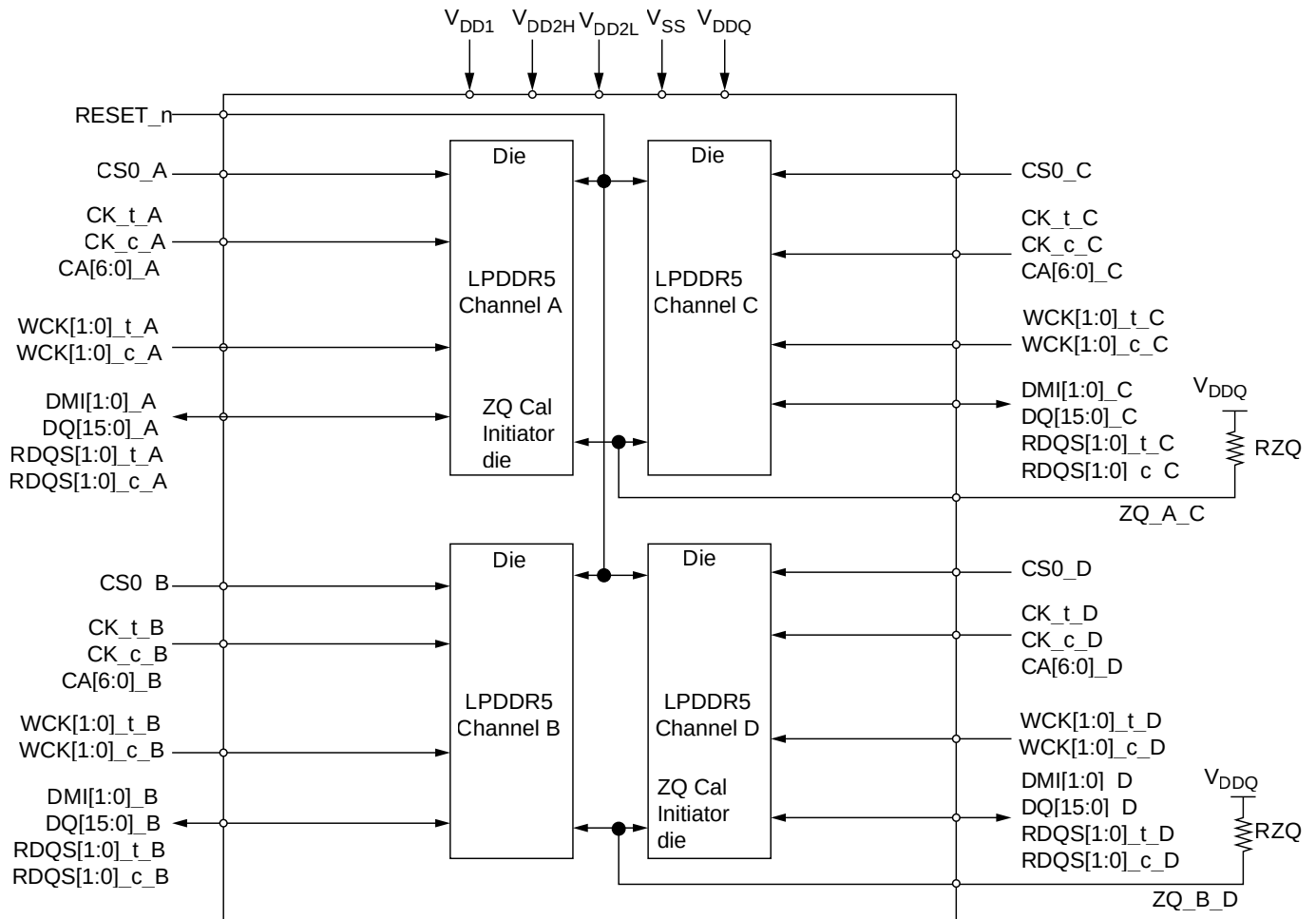




Y6CP LPDDR5X SDRAM Package Block Diagrams

Quad Die, Quad Channel, Single Rank

Figure 8: Quad-Die, Quad-Channel, Single-Rank Package Block Diagram

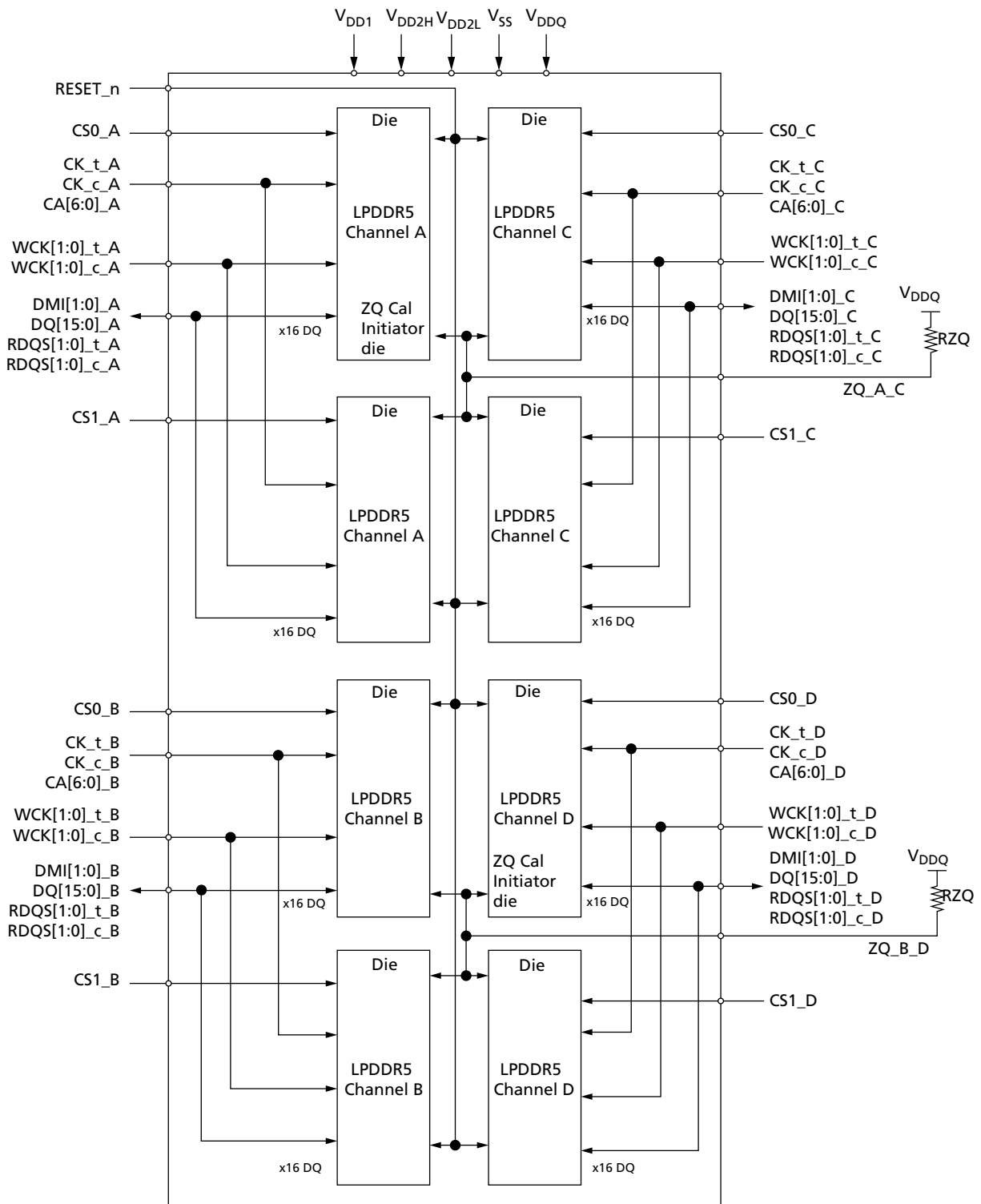




Y6CP LPDDR5X SDRAM Package Block Diagrams

Eight Die, Quad Channel, Dual Rank

Figure 9: Eight-Die, Quad-Channel, Dual-Rank Package Block Diagram





Ball Assignments and Descriptions

315b Dual Channel, 1 Rank, 2 Rank

Table 10: 315-Ball/Pad Descriptions

Symbol	Type	Description
CK_t_[A:B], CK_c_[A:B]	Input	Clock: CK_t and CK_c are differential clock inputs. All double data rate (DDR) command/address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single data rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).
CS0_[A:B], CS1_[A:B]	Input	Chip select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or deep sleep mode where it becomes an asynchronous signal. Each rank (0, 1) has its own CS signals. CS1_[A:B] become NC pins in a single-rank package.
CA[6:0]_[A:B]	Input	Command/address inputs: Provide the command and address inputs according to the command truth table.
WCK[1:0]_t_[A:B], WCK[1:0]_c_[A:B]	Input	Data clock: WCK_t and WCK_c are differential clock inputs used for write data capture and read data output.
DQ[15:0]_[A:B]	I/O	Data input/output: Bidirectional data bus.
RDQS[1:0]_t_[A:B], RDQS[1:0]_c_[A:B]	I/O Output	Read data strobe: RDQS_t and RDQS_c are differential output clock signals used to strobe data during a READ operation. RDQS_t is also used as a parity pin during write link protection enabled. Each byte of data has RDQS_t and RDQS_c signals.
DMI[1:0]_[A:B]	I/O	Data mask inversion: DMI serves multiple functions such as data mask (DM), data bus inversion (DBI), and parity at READ with ECC operation by setting the mode register. DMI is a bidirectional signal and each byte of data has a DMI signal.
ZQ_A	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin should be connected to V _{DDQ} through a 240 ohm $\pm 1\%$ resistor.
V _{DDQ} , V _{DD1} , V _{DD2H} , V _{DD2L}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	Reset: When asserted LOW, the RESET pin resets the die. Reset is an asynchronous signal.
NC	–	No connect: Not internally connected.
RFU	–	Reserved Future Use: Not internally connected.



Y6CP LPDDR5X SDRAM Ball Assignments and Descriptions

Figure 10: 315-Ball Dual-Channel Discrete FBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
A	NC	NC	V _{DDQ}	DMIO_A	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	DMIO_A	V _{DDQ}	NC	NC	A
B	NC	V _{DDQ}	RDQS0_t_A	V _{SS}	DQ4_A	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	DQ12_A	V _{SS}	RDQS1_t_A	V _{DDQ}	NC	B
C	V _{DD1}	DQ1_A	V _{DDQ}	RDQS0_c_A	V _{SS}	DQ5_A	V _{DD2H}	V _{SS}	V _{DD2H}	DQ13_A	V _{SS}	RDQS1_c_A	V _{DDQ}	DQ9_A	V _{DD1}	C
D	DQ0_A	V _{SS}	DQ3_A	V _{DDQ}	WCK0_c_A	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	WCK1_c_A	V _{DDQ}	DQ11_A	V _{SS}	DQ8_A	D
E	V _{SS}	DQ2_A	V _{SS}	WCK0_t_A	V _{DDQ}	DQ6_A	V _{DD2H}	V _{SS}	V _{DD2H}	DQ14_A	V _{DDQ}	WCK1_t_A	V _{SS}	DQ10_A	V _{SS}	E
F	V _{DDQ}	V _{SS}	V _{DDQ}	V _{DDQ}	DQ7_A	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	DQ15_A	V _{DDQ}	V _{DDQ}	V _{SS}	V _{DDQ}	F
G	V _{DDQ}	V _{DDQ}	V _{SS}	CA0_A	V _{SS}	CS1_A	V _{SS}	CA2_A	V _{SS}	CA4_A	V _{SS}	CA6_A	V _{SS}	V _{DDQ}	V _{DDQ}	G
H	RESET_N	V _{DD2L}	V _{SS}	V _{SS}	CA1_A	V _{SS}	CS0_A	V _{SS}	CK_t_A	V _{SS}	CA3_A	V _{SS}	CA5_A	V _{DD2L}	ZQ_A	H
J	V _{SS}	V _{DD2L}	V _{SS}	RFU	V _{DD2H}	RFU	V _{SS}	V _{SS}	CK_c_A	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	J
K	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	K
L	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	L
M	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	M
N	V _{SS}	V _{DD2L}	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	CK_c_B	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	N
P	RFU	V _{DD2L}	CA5_B	V _{SS}	CA3_B	V _{SS}	CK_t_B	V _{SS}	CS0_B	V _{SS}	CA1_B	V _{SS}	V _{SS}	V _{DD2L}	RFU	P
R	V _{DDQ}	V _{DDQ}	V _{SS}	CA6_B	V _{SS}	CA4_B	V _{SS}	CA2_B	V _{SS}	CS1_B	V _{SS}	CA0_B	V _{SS}	V _{DDQ}	V _{DDQ}	R
T	V _{DDQ}	V _{SS}	V _{DDQ}	V _{DDQ}	DQ15_B	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	DQ7_B	V _{DDQ}	V _{DDQ}	V _{SS}	V _{DDQ}	T
U	V _{SS}	DQ10_B	V _{SS}	WCK1_t_B	V _{DDQ}	DQ14_B	V _{DD2H}	V _{SS}	V _{DD2H}	DQ6_B	V _{DDQ}	WCK0_t_B	V _{SS}	DQ2_B	V _{SS}	U
V	DQ8_B	V _{SS}	DQ11_B	V _{DDQ}	WCK1_c_B	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	WCK0_c_B	V _{DDQ}	DQ3_B	V _{SS}	DQ0_B	V
W	V _{DD1}	DQ9_B	V _{DDQ}	RDQS1_c_B	V _{SS}	DQ13_B	V _{DD2H}	V _{SS}	V _{DD2H}	DQ5_B	V _{SS}	RDQS0_c_B	V _{DDQ}	DQ1_B	V _{DD1}	W
Y	NC	V _{DDQ}	RDQS1_t_B	V _{SS}	DQ12_B	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	DQ4_B	V _{SS}	RDQS0_t_B	V _{DDQ}	NC	Y
AA	NC	NC	V _{DDQ}	DMIO_B	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	DMIO_B	V _{DDQ}	NC	NC	AA
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	

Top View (ball down)

V_{SS}
 V_{DD1}
 V_{DD2H}
 V_{DD2L}
 V_{DDQ}
 CK
 RDQS
 WCK
 DQ,DMI
 CA, CS, ZQ, RESET
 NC, RFU



Y6CP LPDDR5X SDRAM Ball Assignments and Descriptions

441-Ball Quad Channel, 1 Rank, 2 Rank

Table 11: 441-Ball/Pad Descriptions

Symbol	Type	Description
CK_t_[A:D], CK_c_[A:D]	Input	Clock: CK_t and CK_c are differential clock inputs. All double data rate (DDR) command/address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single data rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).
CS0_[A:D], CS1_[A:D]	Input	Chip select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or deep sleep mode where it becomes an asynchronous signal. Each rank (0, 1) has its own CS signals. CS1_[A:D] become NC pins in a single-rank package.
CA[6:0]_[A:D]	Input	Command/address inputs: Provide the command and address inputs according to the command truth table.
WCK[1:0]_t_[A:D], WCK[1:0]_c_[A:D]	Input	Data clock: WCK_t and WCK_c are differential clock inputs used for WRITE data capture and READ data output.
DQ[15:0]_[A:D]	I/O	Data input/output: Bidirectional data bus.
RDQS[1:0]_t_[A:D], RDQS[1:0]_c_[A:D]	I/O Output	Read data strobe: RDQS_t and RDQS_c are differential output clock signals used to strobe data during a READ operation. RDQS_t is also used as a parity pin during write link protection enabled. Each byte of data has RDQS_t and RDQS_c signals.
DMI[1:0]_[A:D]	I/O	Data mask inversion: DMI serves multiple functions such as data mask (DM), data bus inversion (DBI), and parity at READ with ECC operation by setting the mode register. DMI is a bidirectional signal and each byte of data has a DMI signal.
ZQ_A_C, ZQ_B_D	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin should be connected to V _{DDQ} through a 240 ohm $\pm 1\%$ resistor.
V _{DDQ} , V _{DD1} , V _{DD2H} , V _{DD2L}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	Reset: When asserted LOW, the RESET pin resets the die. Reset is an asynchronous signal.
NC	–	No connect: Not internally connected.
RFU	–	Reserved Future Use: Not internally connected



Y6CP LPDDR5X SDRAM
Ball Assignments and Descriptions

Figure 11: 441-Ball Quad-Channel FBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
A	V _{SS}	V _{SS}	V _{DD1}	V _{DD2L}	V _{SS}	V _{DD2H}	V _{DD1}	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD1}	V _{DD2L}	V _{SS}	V _{DD2H}	V _{DD1}	V _{SS}	V _{DD2L}	V _{SS}	V _{SS}	A
B	V _{SS}	DQ0_A	V _{SS}	DQ3_A	V _{DD2H}	V _{SS}	DQ11_A	DQ9_A	DQ8_A	V _{SS}	V _{DD2H}	DQ0_C	V _{SS}	DQ3_C	V _{DD2H}	V _{SS}	DQ11_C	DQ9_C	DQ8_C	RFU	V _{SS}	B
C	V _{DD2H}	V _{SS}	DQ2_A	V _{DDQ}	CA0_A	V _{DD2H}	V _{SS}	DQ10_A	V _{DDQ}	V _{DD2H}	V _{SS}	V _{SS}	DQ2_C	V _{DDQ}	CA0_C	V _{DD2H}	V _{SS}	DQ10_C	V _{DDQ}	V _{DD2H}	V _{DD2H}	C
D	V _{SS}	DQ1_A	WCK0_c_A	V _{SS}	CA1_A	CS0_A	V _{DDQ}	V _{SS}	WCK1_t_A	V _{DD2H}	V _{DDQ}	DQ1_C	WCK0_c_C	V _{SS}	CA1_C	CS0_C	V _{DDQ}	V _{SS}	WCK1_t_C	V _{DDQ}	V _{SS}	D
E	V _{DDQ}	RDQS0_c_A	V _{SS}	WCK0_t_A	V _{SS}	CS1_A	V _{SS}	WCK1_c_A	DMI1_A	V _{SS}	V _{DDQ}	RDQS0_c_C	V _{SS}	WCK0_t_C	V _{SS}	CS1_C	V _{SS}	WCK1_c_C	DMI1_C	V _{SS}	V _{DD2H}	E
F	V _{DDQ}	RDQS0_t_A	V _{SS}	V _{DDQ}	V _{SS}	CA2_A	V _{SS}	RDQS1_t_A	V _{SS}	V _{DDQ}	V _{SS}	RDQS0_t_C	V _{SS}	V _{DDQ}	V _{SS}	CA2_C	V _{SS}	RDQS1_t_C	V _{SS}	V _{DDQ}	V _{DD2H}	F
G	V _{SS}	DQ4_A	V _{DDQ}	DMI0_A	RFU	RFU	CA6_A	V _{SS}	RDQS1_c_A	V _{SS}	V _{DDQ}	DMI0_C	V _{DDQ}	DQ4_C	RFU	RFU	CA6_C	V _{SS}	RDQS1_c_C	V _{SS}	V _{SS}	G
H	V _{DD2L}	V _{SS}	DQ5_A	V _{SS}	CK_t_A	V _{SS}	CA5_A	V _{DDQ}	V _{SS}	DQ12_A	V _{SS}	V _{SS}	DQ5_C	V _{SS}	CK_t_C	V _{SS}	CA5_C	V _{DDQ}	V _{SS}	DQ12_C	V _{DD2L}	H
J	V _{DD2H}	DQ6_A	DQ7_A	V _{DD2H}	V _{SS}	CK_c_A	V _{SS}	DQ14_A	DQ13_A	V _{SS}	V _{DD2L}	DQ6_C	DQ7_C	V _{DD2L}	ZQ_A_C	CK_c_C	V _{SS}	DQ14_C	DQ13_C	V _{SS}	V _{DD2H}	J
K	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	CA3_A	CA4_A	V _{DD2L}	V _{SS}	DQ15_A	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	CA3_C	CA4_C	V _{DD2H}	V _{SS}	DQ15_C	V _{SS}	K
L	V _{DD2H}	V _{DD2L}	V _{DD2L}	V _{DD2H}	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	V _{DD2H}	V _{DD2L}	V _{DD2L}	V _{DD2H}	L
M	V _{SS}	DQ15_B	V _{SS}	V _{DD2H}	CA4_B	CA3_B	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	DQ15_D	V _{SS}	V _{DD2L}	CA4_D	CA3_D	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	M
N	V _{DD2H}	V _{SS}	DQ13_B	DQ14_B	V _{SS}	CK_c_B	ZQ_B_D	V _{DD2L}	DQ7_B	DQ6_B	V _{DD2L}	V _{SS}	DQ13_D	DQ14_D	V _{SS}	CK_c_D	V _{SS}	V _{DD2H}	DQ7_D	DQ6_D	V _{DD2H}	N
P	V _{DD2L}	DQ12_B	V _{SS}	V _{DDQ}	CA5_B	V _{SS}	CK_t_B	V _{SS}	DQ5_B	V _{SS}	V _{SS}	DQ12_D	V _{SS}	V _{DDQ}	CA5_D	V _{SS}	CK_t_D	V _{SS}	DQ5_D	V _{SS}	V _{DD2L}	P
R	V _{SS}	V _{SS}	RDQS1_c_B	V _{SS}	CA6_B	RFU	RFU	DQ4_B	V _{DDQ}	DMI0_B	V _{DDQ}	V _{SS}	RDQS1_c_D	V _{SS}	CA6_D	RFU	RFU	DMI0_D	V _{DDQ}	DQ4_D	V _{SS}	R
T	V _{DD2H}	V _{DDQ}	V _{SS}	RDQS1_t_B	V _{SS}	CA2_B	V _{SS}	V _{DDQ}	V _{SS}	RDQS0_t_B	V _{SS}	V _{DDQ}	V _{SS}	RDQS1_t_D	V _{SS}	CA2_D	V _{SS}	V _{DDQ}	V _{SS}	RDQS0_t_D	V _{DDQ}	T
U	V _{DD2H}	V _{SS}	DMI1_B	WCK1_c_B	V _{SS}	CS1_B	V _{SS}	WCK0_t_B	V _{SS}	RDQS0_c_B	V _{DDQ}	V _{SS}	DMI1_D	WCK1_c_D	V _{SS}	CS1_D	V _{SS}	WCK0_t_D	V _{SS}	RDQS0_c_D	V _{DDQ}	U
V	V _{SS}	V _{DDQ}	WCK1_t_B	V _{SS}	V _{DDQ}	CS0_B	CA1_B	V _{SS}	WCK0_c_B	DQ1_B	V _{DDQ}	V _{DD2H}	WCK1_t_D	V _{SS}	V _{DDQ}	CS0_D	CA1_D	V _{SS}	WCK0_c_D	DQ1_D	V _{SS}	V
W	V _{DD2H}	V _{DD2H}	V _{DDQ}	DQ10_B	V _{SS}	V _{DD2H}	CA0_B	V _{DDQ}	DQ2_B	V _{SS}	V _{SS}	V _{DD2H}	V _{DDQ}	DQ10_D	V _{SS}	V _{DD2H}	CA0_D	V _{DDQ}	DQ2_D	V _{SS}	V _{DD2H}	W
Y	V _{SS}	RESET_N	DQ8_B	DQ9_B	DQ11_B	V _{SS}	V _{DD2H}	DQ3_B	V _{SS}	DQ0_B	V _{DD2H}	V _{SS}	DQ8_D	DQ9_D	DQ11_D	V _{SS}	V _{DD2H}	DQ3_D	V _{SS}	DQ0_D	V _{SS}	Y
AA	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	V _{DD1}	V _{DD2H}	V _{SS}	V _{DD2L}	V _{DD1}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	V _{DD1}	V _{DD2H}	V _{SS}	V _{DD2L}	V _{DD1}	V _{SS}	V _{SS}	AA

Top View (ball down)

V _{SS}	V _{DD1}	V _{DD2H}	V _{DD2L}	V _{DDQ}	CK	RDQS	WCK	DQ, DMI	CA, CS, ZQ, RESET	NC
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Y6CP LPDDR5X SDRAM Ball Assignments and Descriptions

561-Ball Quad Channel, 1 Rank, 2 Rank

Table 12: Ball/Pad Descriptions

Symbol	Type	Description
CK_t_[A:D] CK_c_[A:D]	Input	Clock: CK_t and CK_c are differential clock inputs. All double data rate (DDR) command/address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single data rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).
CS0_[A:D], CS1_[A:D]	Input	Chip select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or deep sleep mode where it becomes an asynchronous signal. Each rank (0, 1) has its own CS signals. CS1_[A:D] become NC pins in a single-rank package.
CA[6:0]_[A:D]	Input	Command/address inputs: Provide the command and address inputs according to the command truth table.
WCK[1:0]_t_[A:D] WCK[1:0]_c_[A:D]	Input	Data clock: WCK_t and WCK_c are differential clock inputs used for WRITE data capture and READ data output.
DQ[15:0]_[A:D]	I/O	Data input/output: Bidirectional data bus.
RDQS[1:0]_t_[A:D] RDQS[1:0]_c_[A:D]	I/O Output	Read data strobe: RDQS_t and RDQS_c are differential output clock signals used to strobe data during a READ operation. RDQS_t is also used as a parity pin during write link protection enabled. Each byte of data has RDQS_t and RDQS_c signals.
DMI[1:0]_[A:D]	I/O	Data mask inversion: DMI serves multiple functions such as data mask (DM), data bus inversion (DBI), and parity at READ with ECC operation by setting the mode register. DMI is a bidirectional signal and each byte of data has a DMI signal.
ZQ_A_C, ZQ_B_D	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin should be connected to V _{DDQ} through a 240 ohm $\pm 1\%$ resistor.
V _{DDQ} , V _{DD1} , V _{DD2H} , V _{DD2L}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	Reset: When asserted LOW, the RESET pin resets the die. Reset is an asynchronous signal.
NC	–	No connect: Not internally connected.

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Y6CP LPDDR5X SDRAM
Ball Assignments and Descriptions

Figure 12: 561-Ball Quad-Channel FBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
A		NC	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	NC	NC	A
B	NC	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	NC	B
C	VSS	VSS	DQ0_A	VDDQ	VSS	DMI0_A	DQ5_A	VSS	CS0_A	CSL_A	CA3_A	VSS	DQ13_A	DMI1_A	VSS	VDDQ	DQ8_A	VSS	VSS	C
D	VSS	VDD1	VSS	DQ3_A	WCK0_C_A	VDDQ	VSS	CA0_A	VSS	VSS	VSS	CA6_A	VSS	VDDQ	WCK1_C_A	DQ11_A	VSS	VDD1	VSS	D
E	VSS	VDD2H	DQ1_A	VSS	WCK0_T_A	VSS	DQ6_A	VDDQ	CA1_A	CK_T_A	CA4_A	VDDQ	DQ14_A	VSS	WCK1_T_A	VSS	DQ9_A	VDD2H	VSS	E
F	VSS	VSS	VDDQ	RDQS1_T_A	VDD2H	DQ4_A	VDDQ	VSS	VSS	CK_C_A	VSS	VSS	VDDQ	DQ12_A	VDD2H	RDQS1_T_A	VDDQ	VSS	VSS	F
G	VSS	VDD2L	DQ2_A	RDQS0_C_A	VSS	VDD2H	DQ7_A	VDD2H	CA2_A	VDD2L	CA5_A	VDD2H	DQ15_A	VDD2H	VSS	RDQS1_C_A	DQ10_A	ZQ_A_C	VSS	G
H	VSS	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2L	VSS	H
J	VSS	VDD2L	DQ2_C	RDQS0_C_C	VSS	VDD2H	DQ7_C	VDD2H	CA2_C	VDD2L	CA5_C	VDD2H	DQ15_C	VDD2H	VSS	RDQS1_C_C	DQ10_C	RESET_N	VSS	J
K	VSS	VSS	VDDQ	RDQS0_T_C	VDD2H	DQ4_C	VDDQ	VSS	VSS	CK_C_C	VSS	VSS	VDDQ	DQ12_C	VDD2H	RDQS1_T_C	VDDQ	VSS	VSS	K
L	VSS	VDD2H	DQ1_C	VSS	WCK0_T_C	VSS	DQ6_C	VDDQ	CA1_C	CK_T_C	CA4_C	VDDQ	DQ14_C	VSS	WCK1_T_C	VSS	DQ9_C	VDD2H	VSS	L
M	VSS	VSS	VSS	DQ3_C	WCK0_C_C	VDDQ	VSS	CA0_C	VSS	VSS	VSS	CA6_C	VSS	VDDQ	WCK1_C_C	DQ11_C	VSS	VSS	VSS	M
N	VSS	VDD1	DQ0_C	VDDQ	VSS	DMI0_C	DQ5_C	VSS	CS0_C	CSL_C	CA3_C	VSS	DQ13_C	DMI1_C	VSS	VDDQ	DQ8_C	VDD1	VSS	N
P	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	P
R	VSS	VSS					VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	R
T	VSS	VSS					VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	T
U	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	U
V	VSS	VDD1	DQ8_D	VDDQ	VSS	DMI1_D	DQ13_D	VSS	CA3_D	CSL_D	CS0_D	VSS	DQ5_D	DMI0_D	VSS	VDDQ	DQ0_D	VDD1	VSS	V
W	VSS	VSS	VSS	DQ11_D	WCK1_C_D	VDDQ	VSS	CA6_D	VSS	VSS	VSS	CA0_D	VSS	VDDQ	WCK0_C_D	DQ3_D	VSS	VSS	VSS	W
Y	VSS	VDD2H	DQ9_D	VSS	WCK1_T_D	VSS	DQ14_D	VDDQ	CA4_D	CK_T_D	CA1_D	VDDQ	DQ6_D	VSS	WCK0_T_D	VSS	DQ1_D	VDD2H	VSS	Y
AA	VSS	VSS	VDDQ	RDQS1_T_D	VDD2H	DQ12_D	VDDQ	VSS	VSS	CK_C_D	VSS	VSS	VDDQ	DQ4_D	VDD2H	RDQS0_T_D	VDDQ	VSS	VSS	AA
AB	VSS	VDD2L	DQ10_D	RDQS1_C_D	VSS	VDD2H	DQ15_D	VDD2H	CA5_D	VDD2L	CA2_D	VDD2H	DQ7_D	VDD2H	VSS	RDQS0_C_D	DQ2_D	VSS	VSS	AB
AC	VSS	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2L	VSS	AC
AD	VSS	VDD2L	DQ10_B	RDQS1_C_B	VSS	VDD2H	DQ15_B	VDD2H	CA5_B	VDD2L	CA2_B	VDD2H	DQ7_B	VDD2H	VSS	RDQS0_C_B	DQ2_B	ZQ_B_D	VSS	AD
AE	VSS	VSS	VDDQ	RDQS1_T_B	VDD2H	DQ12_B	VDDQ	VSS	VSS	CK_C_B	VSS	VSS	VDDQ	DQ4_B	VDD2H	RDQS0_T_B	VDDQ	VSS	VSS	AE
AF	VSS	VDD2H	DQ9_B	VSS	WCK1_T_B	VSS	DQ14_B	VDDQ	CA4_B	CK_T_B	CA1_B	VDDQ	DQ6_B	VSS	WCK0_T_B	VSS	DQ1_B	VDD2H	VSS	AF
AG	VSS	VDD1	VSS	DQ11_B	WCK1_C_B	VDDQ	VSS	CA6_B	VSS	VSS	VSS	CA0_B	VSS	VDDQ	WCK0_C_B	DQ3_B	VSS	VDD1	VSS	AG
AH	VSS	VSS	DQ8_B	VDDQ	VSS	DMI1_B	DQ13_B	VSS	CA3_B	CSL_B	CS0_B	VSS	DQ5_B	DMI0_B	VSS	VDDQ	DQ0_B	VSS	VSS	AH
AJ	NC	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	NC	AJ
AK	NC	NC	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	NC	NC	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	

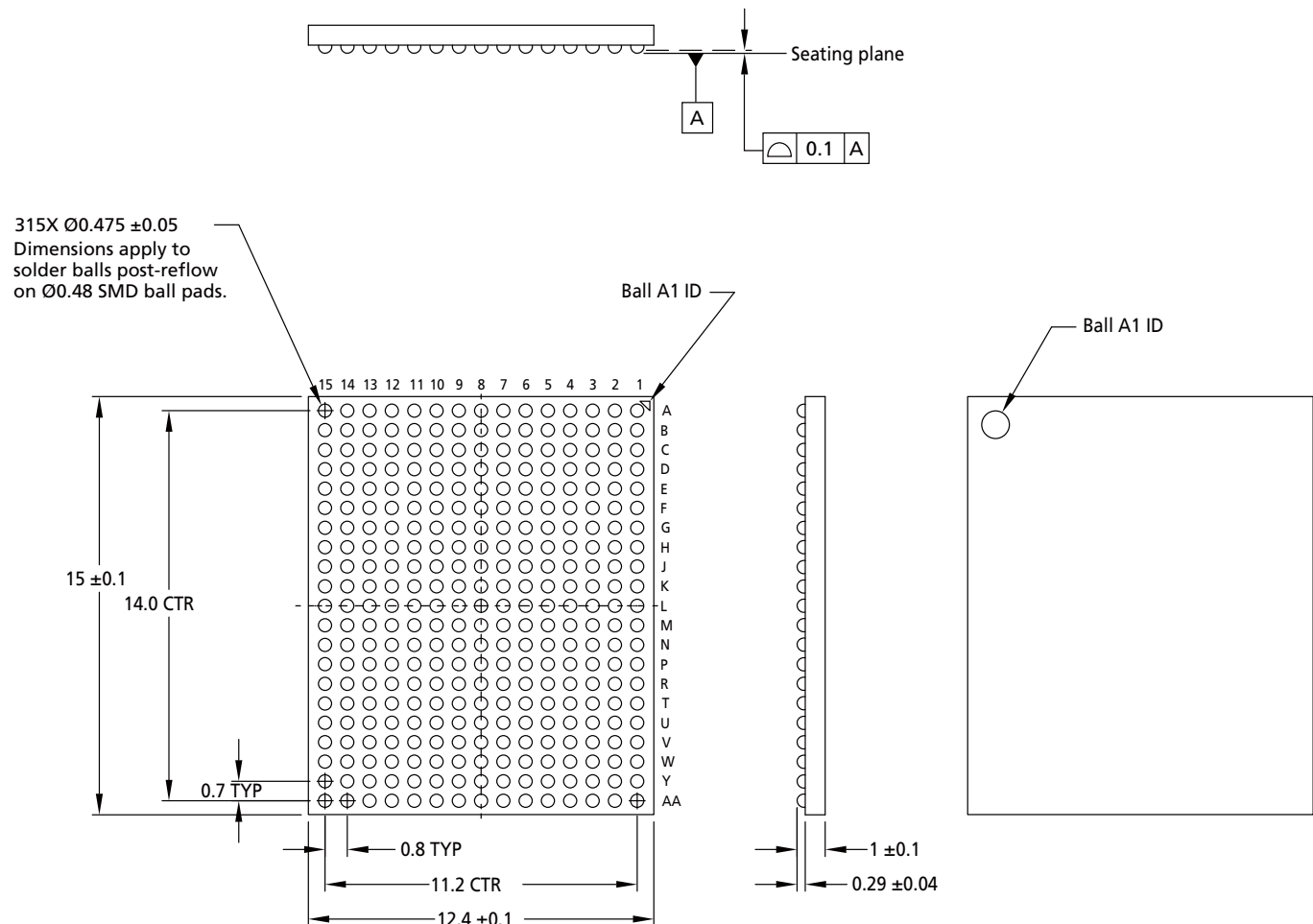


Y6CP LPDDR5X SDRAM Package Dimensions

Package Dimensions

315-Ball Package (Package Code: DS)

Figure 13: 315-Ball TFBGA – 12.4mm (TYP) × 15.0mm (TYP) × 1.1mm (MAX)



Notes: 1. All dimensions are in millimeters.

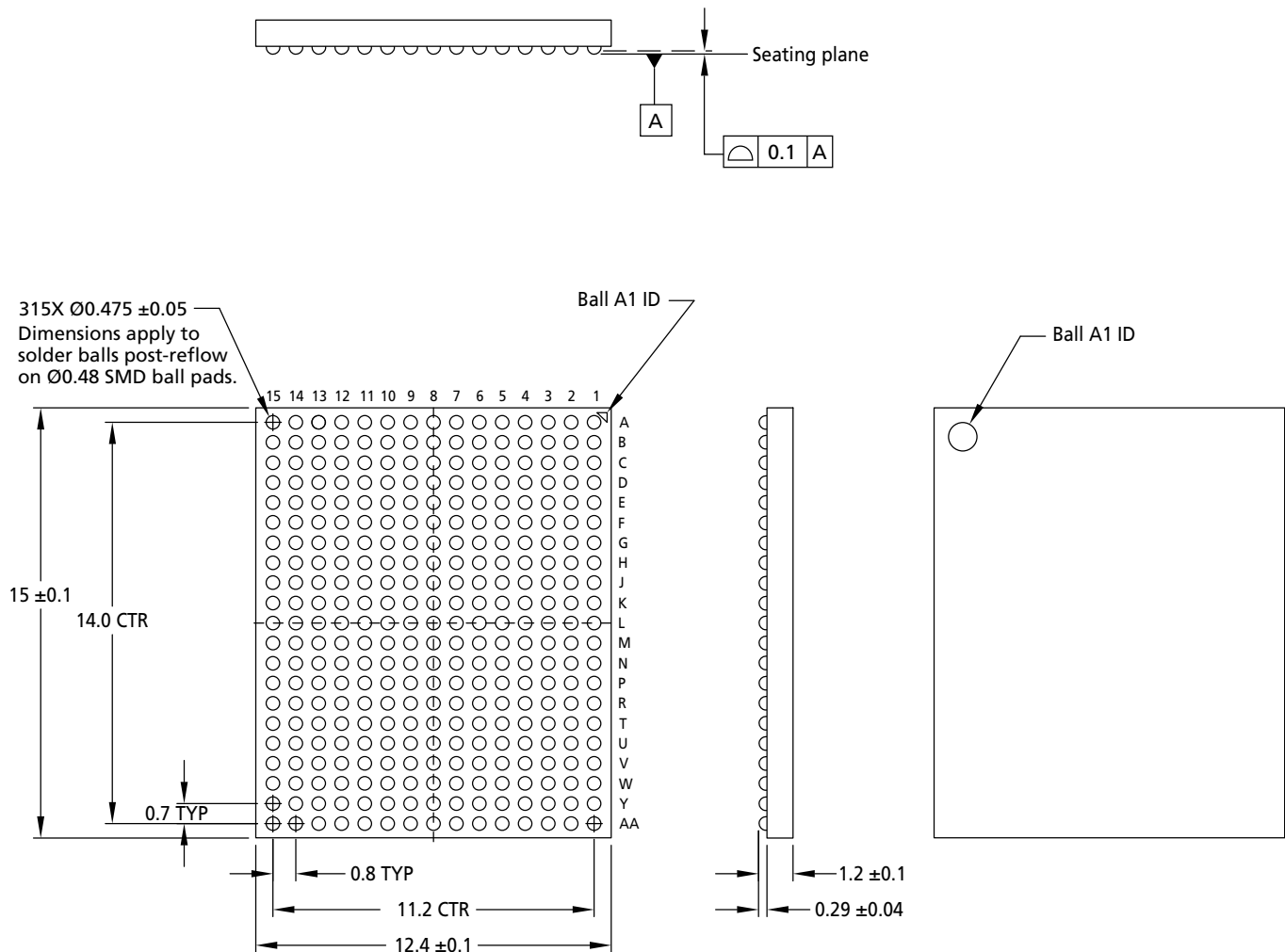
2. Solder ball composition: SACQ with CuOSP pads (Sn-4Ag-0.5Cu-3Bi-0.05Ni)



Y6CP LPDDR5X SDRAM Package Dimensions

315-Ball Package (Package Code: DV)

Figure 14: 315-Ball LFBGA – 12.4mm (TYP) × 15.0mm (TYP) × 1.3mm (MAX)



Notes: 1. All dimensions are in millimeters.

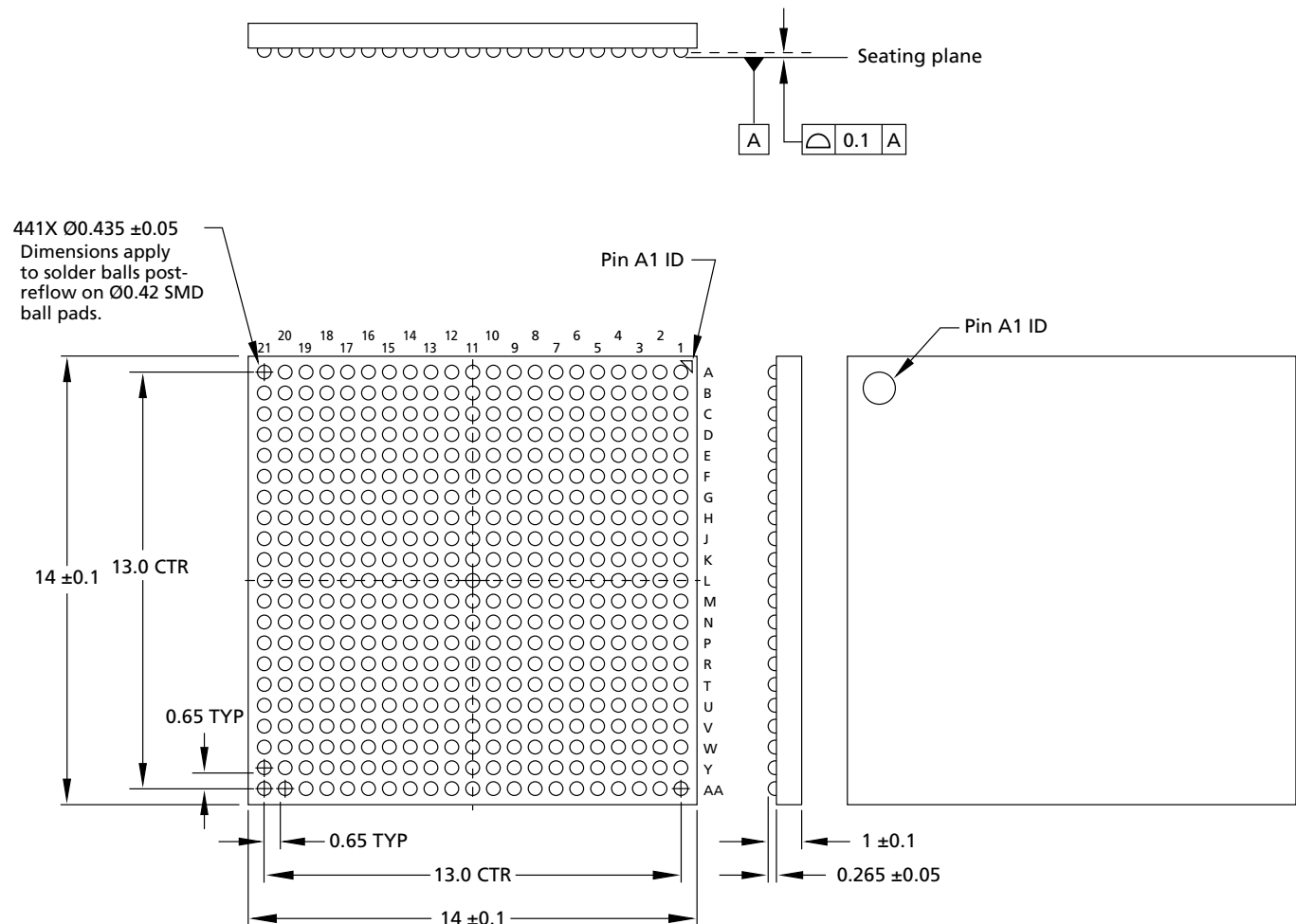
2. Solder ball composition: SACQ with CuOSP pads (Sn-4Ag-0.5Cu-3Bi-0.05Ni)



Y6CP LPDDR5X SDRAM Package Dimensions

441-Ball Package (Package Code: EK)

Figure 15: 441-Ball TFBGA – 14.0mm (TYP) x 14.0mm (TYP) x 1.1mm (MAX)

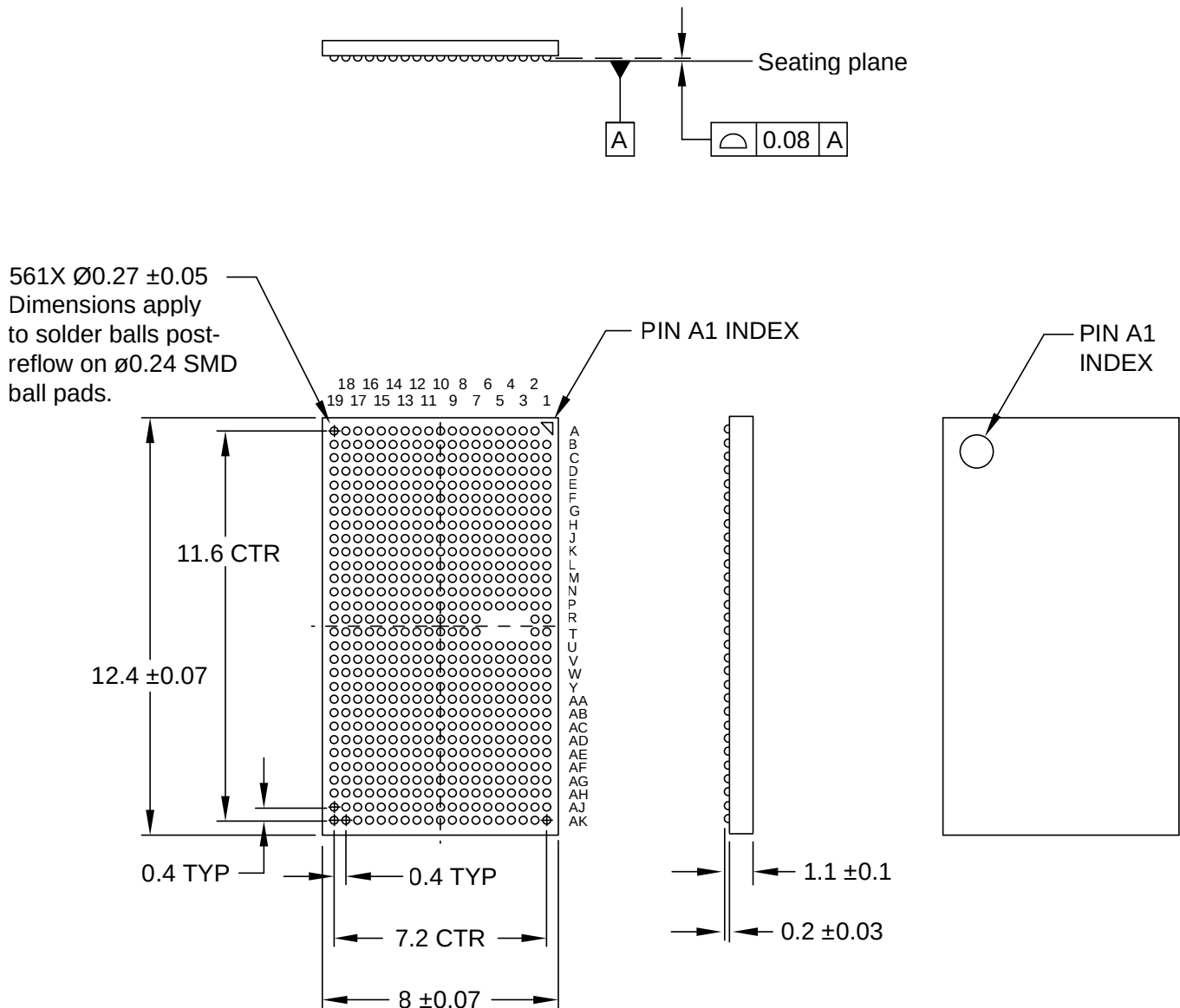


Notes: 1. All dimensions are in millimeters.

2. Solder ball composition: SACQ with CuOSP pads (Sn-4Ag-0.5Cu-3Bi-0.05Ni).

561-Ball Package (Package Code: CZ)

Figure 16: 561-Ball TFBGA – 8.0mm (TYP) × 12.4mm (TYP) × 1.2mm (MAX)



Notes: 1. All dimensions are in millimeters.

2. Solder ball composition: SACQ with CuOSP pads (Sn- 4Ag-0.5Cu-3Bi-0.05Ni)


Y6CP LPDDR5X SDRAM
Package Dimensions
Package Thermal Impedance
Table 13: Package Thermal Impedance Characteristics

Die Revision	Package	Parameter	Symbol	Value	Unit
Rev B	315-ball, DDP, package code DS	Junction-to-case (TOP)	Θ_{JC}	2.8	°C/W
		Junction-to-board	Θ_{JB}	11.4	°C/W
	315-ball, QDP, package code DS	Junction-to-case (TOP)	Θ_{JC}	3.3	°C/W
		Junction-to-board	Θ_{JB}	12.8	°C/W
	315-ball, 8DP, package code DV	Junction-to-case (TOP)	Θ_{JC}	3.3	°C/W
		Junction-to-board	Θ_{JB}	8.6	°C/W
	441-ball, QDP, package code EK	Junction-to-case (TOP)	Θ_{JC}	2.3	°C/W
		Junction-to-board	Θ_{JB}	8.7	°C/W
	441-ball, 8DP, package code EK	Junction-to-case (TOP)	Θ_{JC}	2.3	°C/W
		Junction-to-board	Θ_{JB}	8.4	°C/W
	561-ball, QDP, package code CZ	Junction-to-case (TOP)	Θ_{JC}	5.0	°C/W
		Junction-to-board	Θ_{JB}	13.6	°C/W
	561-ball, 8DP, package code CZ	Junction-to-case (TOP)	Θ_{JC}	4.6	°C/W
		Junction-to-board	Θ_{JB}	14.1	°C/W



Y6CP LPDDR5X SDRAM Product-Specific Mode Register Definition

Product-Specific Mode Register Definition

Table 14: Mode Register Contents

Mode Register	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR0	Per-pin DFE	Pre-emphasis	Unified NT ODT behavior mode	DMI output behavior mode	Optimized refresh mode	Enhanced WCK always-on mode	Latency mode	NT ODT timing mode
	OP[0] = 1b: Device supports different NT ODT latency for DQ and RDQS							
	OP[1] = 0b: Device supports x16 mode latency for 1536M32, 3G32, 1536M64, 3G64 OP[1] = 1b: Device supports x8 mode latency for 6G32							
	OP[2] = 1b: Device supports enhanced WCK always-on mode							
	OP[3] = 1b: Device supports optimized refresh mode							
	OP[4] = 1b: Device supports both DMI behavior mode 1 and 2 and mode selection							
	OP[5] = 1b: The NT ODT behavior follows the unified NT ODT behavior							
	OP[6] = 1b: Device supports pre-emphasis mode							
OP[7] = 1b: Device supports per-pin DFE								
MR1						DRFM support ⁶	ARFM support ³	CS ODT OP support
	OP[0] = 1b: Device supports CS ODT behavior OP							
	OP[1] = 1b: Device supports ARFM							
	OP[2] = 1b: Device supports DRFM							
MR3				BK/BG ORG				
	OP[4:3] = 00b: BG mode supported							
	01b: 8B mode not supported							
	10b: 16B Mode supported							
	11b: RFU							
MR4				Refresh Multiplier ⁷				
	OP[4:0]: Refer to General LPDDR5/LPDDR5X Specification 1 for mode register definitions.							
MR5	Manufacturer ID							
	1111 1111b: Micron							
MR6	Revision ID1							
	0001 1001b							
MR8	I/O width		Density				Type	
	OP[7:6] = 00b: x16 for 1536M32, 3G32, 1536M64, 3G64 OP[7:6] = 01b: x8 for 6G32		OP[5:2] = 0111b: 24Gb				OP[1:0] = 11b: LPDDR5X (up to 10.7 Gb/s) 10b: LPDDR5X (up to 9600 Mb/s)	
MR19			WCK2DQ OSC FM					
	OP[5] = 1b: WCK2DQ OSC FM supported							



Y6CP LPDDR5X SDRAM

Product-Specific Mode Register Definition

Table 14: Mode Register Contents (Continued)

Mode Register	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR21	WXS				ODTD-CSFS	WXFS	RDCFS	WDCFS
	OP[0] = 1b: WRITE DATA COPY function supported							
	OP[1] = 1b: READ DATA COPY function supported							
	OP[2] = 1b: WRITE X function supported							
	OP[3] = 1b: Device ODTD-CS is supported							
	OP[7] = 1b: Data to be written can be selected with 0 and 1							
MR24	DFES				Read DCA			
	OP[3] = 1b: Device supports Read DCA							
	OP[7] = 1b: Device supports DFE (See Note 4)							
MR26		RDQSTFS						
	OP[6] = 1b: Read/write-based RDQS_t TRAINING function supported							
MR27	RAAMULT		RAAIMT					RFM
	OP[0] = 1b: RFM is required							
	OP[5:1] = 00101b: 64							
	OP[7:6] = 01b: 4X							
MR41					E-DVFSC ODT	DVFSC/E-DVFSC Support		
	OP[2:1] = 10b: Both DVFSC and Enhanced DVFSC Mode supported							
	OP[3] = 1b: Enhanced DVFSC ODT option is supported							
MR43		SBEC rule						
	OP[6] = 1b: Simultaneous SBE on each DQ byte and DMI are independently counted							
MR57					RFMSB		RAADEC	
	OP[1:0] = 10b: 2 × RAAIMT							
	OP[3:2] = 01b: = Does support single-bank mode							
MR75								BRCS
	OP[0] = 1b: BRC is supported							
MR63-MR74, MR76 -MR164	Reserved MR bits MR63 through MR74 and MR76 through MR164 are RFU by JEDEC standard and should not be accessed by user.							

- Notes: 1. The contents of mode registers described here reflect information specific to each die in these packages.
 2. Refer to General LPDDR5/LPDDR5X Specification 1 for mode registers not described here.
 3. Refer to General LPDDR5/LPDDR5X Specification 3 for feature description not described here.
 4. Device supports 7-step DFE.
 5. Device type only supports Bank Group and 16B modes.
 6. Refer to General LPDDR5/LPDDR5X Specification 3 for DRFM details not described here
 7. MR4:OP[4:0] output from each die in a defined package shall be used to determine the required package level refresh rate. Any other method used to determine a valid refresh rate, places ownership and responsibility of the correct refresh rate on the system developer or end user. T_{OPER} (case surface temperature) shall be used to determine whether operating temperature requirements are being met. Any other method used to determine operating temperature, places ownership and responsibility of the correct operating temperature on the system developer or end user.


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
I_{DD} Parameters

Refer to the I_{DD} Specification Parameters and Test Conditions section in General LPDDR5 Specifications 2 for detailed conditions.

Table 15: WT I_{DD} Parameters – Single Die

Symbol	Supply	Speed Grade				Unit	Note
		x8 Mode		x16 Mode			
		9600 Mb/s	10,667 Mb/s	9600 Mb/s	10,667 Mb/s		
I _{DD01}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD02H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD02L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD0Q}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2P1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2P2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2P2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2PQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2PS1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2PS2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2PS2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2PSQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2N1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2N2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2N2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2NQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2NS1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2NS2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2NS2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2NSQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD3P1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD3P2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD3P2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD3PQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD3PS1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD3PS2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD3PS2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD3PSQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD3N1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD3N2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD3N2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD3NQ}	V _{DDQ}	TBD	TBD	TBD	TBD		


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
Table 15: WT I_{DD} Parameters – Single Die (Continued)

Symbol	Supply	Speed Grade				Unit	Note
		x8 Mode		x16 Mode			
		9600 Mb/s	10,667 Mb/s	9600 Mb/s	10,667 Mb/s		
I _{DD4R1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	3, 4, 6
I _{DD4R2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD4R2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD4RQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD4W1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	3, 6
I _{DD4W2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD4W2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD4WQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD51}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD52H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD52L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD5Q}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD5AB1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD5AB2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD5AB2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD5ABQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD5PB1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD5PB2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD5PB2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD5PBQ}	V _{DDQ}	TBD	TBD	TBD	TBD		

Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.

2. BG mode enabled. DVFSC and DVFSQ disabled.

3. BL = 16, DBI disabled.

4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C

5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C

6. IDD4R2 and IDD4W2 have implemented a more stringent pattern than required by JEDEC and therefore these IDD values are slightly higher than if the JEDEC pattern was used.

7. Notes 1, 2, and 5 apply to entire table.


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
Table 16: WT I_{DD} Parameters – With Enhanced DVFS Enabled - Single Die

Symbol	Supply	Speed Grade		Unit	Note
		x8 Mode	x16 Mode		
		3200 Mb/s	3200 Mb/s		
I _{DD4R-DVFS} C1	V _{DD1}	TBD	TBD	mA	3, 4, 6
I _{DD4R-DVFS} C2H	V _{DD2H}	TBD	TBD		
I _{DD4R-DVFS} C2L	V _{DD2L}	TBD	TBD		
I _{DD4R-DVFS} CQ	V _{DDQ}	TBD	TBD		
I _{DD4W-DVFS} C1	V _{DD1}	TBD	TBD	mA	3, 6
I _{DD4W-DVFS} C2H	V _{DD2H}	TBD	TBD		
I _{DD4W-DVFS} C2L	V _{DD2L}	TBD	TBD		
I _{DD4W-DVFS} CQ	V _{DDQ}	TBD	TBD		
I _{DD5ED} 1	V _{DD1}	TBD	TBD	mA	
I _{DD5ED} 2H	V _{DD2H}	TBD	TBD		
I _{DD5ED} 2L	V _{DD2L}	TBD	TBD		
I _{DD5ED} Q	V _{DDQ}	TBD	TBD		
I _{DD5ABED} 1	V _{DD1}	TBD	TBD	mA	
I _{DD5ABED} 2H	V _{DD2H}	TBD	TBD		
I _{DD5ABED} 2L	V _{DD2L}	TBD	TBD		
I _{DD5ABED} Q	V _{DDQ}	TBD	TBD		
I _{DD5PBED} 1	V _{DD1}	TBD	TBD	mA	
I _{DD5PBED} 2H	V _{DD2H}	TBD	TBD		
I _{DD5PBED} 2L	V _{DD2L}	TBD	TBD		
I _{DD5PBED} Q	V _{DDQ}	TBD	TBD		

- Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.
2. Enhanced DVFS enabled, DVFSQ enabled, 3200 Mb/s, 16B mode
3. BL = 16, DBI disabled.
4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C
5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C
6. IDD4R2 and IDD4W2 have implemented a more stringent pattern than required by JEDEC and therefore these IDD values are slightly higher than if the JEDEC pattern was used.
7. Notes 1, 2, and 5 apply to entire table.


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
Table 17: WT Full-Array Power-Down Self Refresh Current – Single Die

Temperature	Symbol	Supply	Value	Unit	
25°C	I _{DD61}	V _{DD1}	TBD	mA	
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD	uA	4
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD	uA	4
	I _{DD6DSQ}	V _{DDQ}	TBD		
45°C	I _{DD61}	V _{DD1}	TBD	mA	
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD	uA	4
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD	uA	4
	I _{DD6DSQ}	V _{DDQ}	TBD		
65°C	I _{DD61}	V _{DD1}	TBD	mA	
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD	uA	4
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD	uA	4
	I _{DD6DSQ}	V _{DDQ}	TBD		
85°C	I _{DD61}	V _{DD1}	TBD	mA	
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD	mA	4
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD	mA	4
	I _{DD6DSQ}	V _{DDQ}	TBD		

- Notes: 1. I_{DD625/45/65°C} is the typical value in the distribution with nominal V_{DD} and a reference-only value. I_{DD685°C} is the maximum I_{DD} guaranteed value considering the worst-case conditions of process, temperature, and voltage.
2. DVFS_C and DVFS_Q disabled.
3. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C
4. While entering and exiting Self-Refresh modes, all defined/used supply rails (VDD1, VDD2H, VDD2L, VDDQ) are required to be at valid levels. After the Self-Refresh with Power down mode entrance commands are completed and t_{ESPD} timing requirement has been satisfied, the V_{DDQ} power rail may be turned off by the controller.


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
Table 18: IT I_{DD} Parameters – Single Die

Symbol	Supply	Speed Grade				Unit	Note
		x8 Mode		x16 Mode			
		9600 Mb/s	10,667 Mb/s	9600 Mb/s	10,667 Mb/s		
I _{DD01}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD02H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD02L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD0Q}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2P1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2P2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2P2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2PQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2PS1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2PS2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2PS2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2PSQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2N1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2N2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2N2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2NQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD2NS1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD2NS2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD2NS2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD2NSQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD3P1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD3P2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD3P2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD3PQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD3PS1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD3PS2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD3PS2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD3PSQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD3N1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD3N2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD3N2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD3NQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD4R1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	3, 4, 6
I _{DD4R2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD4R2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD4RQ}	V _{DDQ}	TBD	TBD	TBD	TBD		


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
Table 18: IT I_{DD} Parameters – Single Die (Continued)

Symbol	Supply	Speed Grade				Unit	Note
		x8 Mode		x16 Mode			
		9600 Mb/s	10,667 Mb/s	9600 Mb/s	10,667 Mb/s		
I _{DD4W1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	3, 6
I _{DD4W2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD4W2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD4WQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD51}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD52H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD52L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD5Q}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD5AB1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD5AB2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD5AB2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD5ABQ}	V _{DDQ}	TBD	TBD	TBD	TBD		
I _{DD5PB1}	V _{DD1}	TBD	TBD	TBD	TBD	mA	
I _{DD5PB2H}	V _{DD2H}	TBD	TBD	TBD	TBD		
I _{DD5PB2L}	V _{DD2L}	TBD	TBD	TBD	TBD		
I _{DD5PBQ}	V _{DDQ}	TBD	TBD	TBD	TBD		

- Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.
2. BG mode enabled. DVFS, Enhanced DVFS and DVFSQ disabled.
3. BL = 16, DBI disabled.
4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C
5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V (DVFSQ disabled)/0.27–0.37V (DVFSQ enabled); T_C = –40°C to +95°C
6. IDD4R2 and IDD4W2 have implemented a more stringent pattern than required by JEDEC and therefore these IDD values are slightly higher than if the JEDEC pattern was used.
7. Notes 1, 2, and 5 apply to entire table.


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
Table 19: IT I_{DD} Parameters – With Enhanced DVFS Enabled - Single Die

Symbol	Supply	Speed Grade		Unit	Note
		x8 Mode	x16 Mode		
		3200 Mb/s	3200 Mb/s		
I _{DD4R-DVFS} C1	V _{DD1}	TBD	TBD	mA	3, 4, 6
I _{DD4R-DVFS} C2H	V _{DD2H}	TBD	TBD		
I _{DD4R-DVFS} C2L	V _{DD2L}	TBD	TBD		
I _{DD4R-DVFS} CQ	V _{DDQ}	TBD	TBD		
I _{DD4W-DVFS} C1	V _{DD1}	TBD	TBD	mA	3, 6
I _{DD4W-DVFS} C2H	V _{DD2H}	TBD	TBD		
I _{DD4W-DVFS} C2L	V _{DD2L}	TBD	TBD		
I _{DD4W-DVFS} CQ	V _{DDQ}	TBD	TBD		
I _{DD5ED} 1	V _{DD1}	TBD	TBD	mA	
I _{DD5ED} 2H	V _{DD2H}	TBD	TBD		
I _{DD5ED} 2L	V _{DD2L}	TBD	TBD		
I _{DD5ED} Q	V _{DDQ}	TBD	TBD		
I _{DD5ABED} 1	V _{DD1}	TBD	TBD	mA	
I _{DD5ABED} 2H	V _{DD2H}	TBD	TBD		
I _{DD5ABED} 2L	V _{DD2L}	TBD	TBD		
I _{DD5ABED} Q	V _{DDQ}	TBD	TBD		
I _{DD5PBED} 1	V _{DD1}	TBD	TBD	mA	
I _{DD5PBED} 2H	V _{DD2H}	TBD	TBD		
I _{DD5PBED} 2L	V _{DD2L}	TBD	TBD		
I _{DD5PBED} Q	V _{DDQ}	TBD	TBD		

- Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.
2. Enhanced DVFS enabled, DVFSQ enabled, 3200 Mb/s, 16B mode
3. BL = 16, DBI disabled.
4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C
5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C
6. IDD4R2 and IDD4W2 have implemented a more stringent pattern than required by JEDEC and therefore these IDD values are slightly higher than if the JEDEC pattern was used.
7. Notes 1, 2, and 5 apply to entire table.


Y6CP LPDDR5X SDRAM
I_{DD} Parameters
Table 20: IT Full-Array Power-Down Self Refresh Current – Single Die

Temperature	Symbol	Supply	Value	Unit	Notes
25°C	I _{DD61}	V _{DD1}	TBD	mA	4
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD		
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	4
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD		
	I _{DD6DSQ}	V _{DDQ}	TBD		
45°C	I _{DD61}	V _{DD1}	TBD	mA	4
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD		
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	4
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD		
	I _{DD6DSQ}	V _{DDQ}	TBD		
65°C	I _{DD61}	V _{DD1}	TBD	mA	4
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD		
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	4
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD		
	I _{DD6DSQ}	V _{DDQ}	TBD		
85°C	I _{DD61}	V _{DD1}	TBD	mA	4
	I _{DD62H}	V _{DD2H}	TBD		
	I _{DD62L}	V _{DD2L}	TBD		
	I _{DD6Q}	V _{DDQ}	TBD		
	I _{DD6DS1}	V _{DD1}	TBD	mA	4
	I _{DD6DS2H}	V _{DD2H}	TBD		
	I _{DD6DS2L}	V _{DD2L}	TBD		
	I _{DD6DSQ}	V _{DDQ}	TBD		

- Notes: 1. I_{DD625/45/65°C} is the typical value in the distribution with nominal V_{DD} and a reference-only value. I_{DD685°C} is the maximum I_{DD} guaranteed value considering the worst-case conditions of process, temperature, and voltage.
2. DVFSC and DVFSQ disabled.
3. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –40°C to +95°C
4. While entering and exiting self-refresh modes, all defined/used supply rails (V_{DD1}, V_{DD2H}, V_{DD2L}, V_{DDQ}) are required to be at valid levels. After the Self-Refresh with Power down mode entrance commands are completed and t_{ESPD} timing requirement has been satisfied, the V_{DDQ} power rail may be turned off by the controller.



Revision History

Rev. E – 11/2025

- Added Package Thermal Impedance values
- Removed 8533 Mb/s information and add 10,667 Mb/s information
- Removed 496b information

Rev. D – 08/2025

- Updated tolerance of 561b CZ package

Rev. C – 06/2025

- Added 496-Ball GK package

Rev. B – 04/2025

- Added note to thermal impedance page

Rev. A – 06/2024

- Initial Preliminary release

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